








Advanced Time Series Analysis and Forecasting

Chapter 8: Advanced volatility modelling: realised measures, HAR and multivariate GARCH



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/3)

- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARMA**: AutoRegressive Moving Average (model)
- **BEKK**: Baba–Engle–Kraft–Kroner (multivariate GARCH model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNHLS**: Barndorff-Nielsen–Hansen–Lunde–Shephard
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BV**: Bipower Variation
- **CAC**: Cotation Assistée en Continu (CAC 40 index, Paris) — the CAC 40 index of the Paris bourse
- **CCC**: Constant Conditional Correlation (model)
- **CLT**: Central Limit Theorem
- **DCC**: Dynamic Conditional Correlation
- **DCC-NL**: DCC with Nonlinear shrinkage of the target (Engle–Ledoit–Wolf 2019)
- **DM**: Diebold–Mariano (test)
- **EGARCH**: Exponential GARCH
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **FDR**: False Discovery Rate

Acronyms used in this material (2/3)

- **FRED**: Federal Reserve Economic Data
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-MIDAS**: GARCH with a MIxed DAta Sampling long-run component (Engle–Ghysels–Sohn 2013)
- **GARCH-X**: GARCH with eXogenous variables
- **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- **GMM**: Generalized Method of Moments
- **GMV**: Global Minimum-Variance (portfolio)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HAR-CJ**: HAR with Continuous and Jump components
- **HARQ**: HAR with realised Quarticity
- **HEAVY**: High-frEquency-bAsed VolatilitY (models)
- **IGARCH**: Integrated GARCH
- **IQ**: Integrated Quarticity
- **LLM**: Large Language Model
- **MAE**: Mean Absolute Error
- **MFM**: Modelling Financial Markets
- **MIDAS**: MIxed DAta Sampling (regression)
- **ML**: Machine Learning

Acronyms used in this material (3/3)

- **MSE**: Mean Squared Error
- **MSTR**: MicroStrategy (Strategy Inc.), NASDAQ ticker
- **OLS**: Ordinary Least Squares
- **OPG**: Outer Product of the Gradient (scores)
- **PPI**: Producer Price Index
- **QLIKE**: Quasi-LIKElihood loss
- **QML**: Quasi-Maximum Likelihood
- **QMLE**: Quasi-Maximum Likelihood Estimator
- **QV**: Quadratic Variation
- **RGARCH**: Realized GARCH
- **RK**: Realised Kernel
- **RMSE**: Root Mean Squared Error
- **RQ**: Realised Quarticity
- **RV**: Realised Variance
- **SV**: Stochastic Volatility
- **TSA**: Time Series Analysis
- **TSLA**: Tesla Inc., NASDAQ ticker
- **TSRV**: Two-Scales Realised Variance
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VEC**: VEC multivariate GARCH (half-vectorised covariance)
- **VIX**: CBOE Volatility IndeX
- **VR**: Variance Ratio

Today's question and route

- **Question**: how precisely can we measure, model and forecast a variance that is never observed?
  - and which of our tools still work when the data are noisy, jumpy or high-dimensional?
  - two routes to volatility: a parametric filter of daily returns (GARCH) and a nonparametric measurement from intraday prices (realised measures); the frontier combines them
- **Route** of the chapter
  - quasi-maximum likelihood for GARCH and robust inference; long-run components: component GARCH and GARCH-MIDAS with macroeconomic drivers
  - realised measures: asymptotics of realised variance, microstructure noise, realised kernels, bipower variation and jump tests
  - forecasting with realised measures: HAR, HARQ, Realized GARCH, HEAVY; robust loss functions
  - multivariate: BEKK, DCC and cDCC, the curse of dimensionality and DCC-NL, realised covariance
- We build on TSA, Chapter 5 (GARCH, GJR, EGARCH, news impact, QLIKE) and TSA, Chapter 14 (DCC basics)
  - MFM, Chapters 5–9 apply these tools to market risk; Seminar 8 comes before this lecture

Learning outcomes

- State the conditions under which Gaussian QML for GARCH is consistent and asymptotically normal, and compute Bollerslev–Wooldridge standard errors
- Separate short-run from long-run volatility with component GARCH and GARCH-MIDAS, and judge whether a macroeconomic driver matters
- Derive the limit theory of realised variance, explain the effect of noise and jumps, and choose among RV, realised kernels and bipower variation
- Forecast volatility with HAR, HARQ and Realized GARCH, and compare forecasts with losses that are robust to a noisy proxy
- Estimate DCC and cDCC models, explain why large covariance matrices need shrinkage, and measure correlation from intraday data

Reading, data and tools

- Backbone: Francq and Zakoïan (2019); Andersen, Bollerslev, Diebold and Labys (2003); Barndorff-Nielsen and Shephard (2002); Barndorff-Nielsen, Hansen, Lunde and Shephard (2008)
  - forecasting and the multivariate case: Corsi (2009); Hansen, Huang and Shek (2012); Engle (2002); Patton (2011)
  - case studies: Engle, Ghysels and Sohn (2013) (GARCH-MIDAS), Bollerslev, Patton and Quaedvlieg (2016) (HARQ), Hansen, Huang and Shek (2012) (Realized GARCH), Engle, Ledoit and Wolf (2019) (DCC-NL)
- Python Quantlets of this chapter: Quantlets/Ch_08
  - QML with sandwich standard errors, component GARCH, GARCH-MIDAS, realised kernels, jump tests, HAR/HARQ, Realized GARCH, DCC/cDCC and nonlinear shrinkage written out in `numpy`
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
Daily realised measures of six equity indices (5-minute RV, BV, realised kernel, open-to-close return), 2000–2022	Heber, Lunde, Shephard and Sheppard (2009), v0.3	HAR, RGARCH
Bitcoin and Ether: one-minute prices 2018–2026, one-second prices August 2026	Binance public data (data.binance.vision); measures computed by us	noise, jumps, HARQ
S&P 500, DAX, BET, Bitcoin, 14 US stocks and 3 equity ETFs, daily	EODHD	QML, MIDAS, DCC
EUR/RON reference rate	BNR	QML
US industrial production, PPI finished goods (monthly)	FRED (INDPRO, WPSFD49207)	GARCH-MIDAS

- The Oxford-Man library ended in February 2022 (archived copy, cited as its terms require); Binance one-minute data extend the realised measures to 18 September 2026; a calibrated simulation gives the theory a known truth
- RV: realised variance; BV: bipower variation; PPI: producer price index

A short history of volatility modelling



New York Stock Exchange, 1908

Photo: Helen D. Van Eaton, Leslie's Monthly Magazine (1908); public domain; Wikimedia Commons

- 1982–1986: ARCH and GARCH Engle (1982), Bollerslev (1986): volatility as a filtered conditional variance of daily returns
- 1992–2004: QML theory Bollerslev and Wooldridge (1992), Lee and Hansen (1994), Lumsdaine (1996), Francq and Zakoïan (2004); components Engle and Lee (1999); DCC Engle (2002)
- 1998–2008: realised variance Andersen and Bollerslev (1998), Andersen, Bollerslev, Diebold and Labys (2003), Barndorff-Nielsen and Shephard (2002); bipower variation Barndorff-Nielsen and Shephard (2004); noise and kernels Zhang, Mykland and Aït-Sahalia (2005), Barndorff-Nielsen, Hansen, Lunde and Shephard (2008)
- 2009–2019: HAR Corsi (2009), Realized GARCH Hansen, Huang and Shek (2012), GARCH-MIDAS Engle, Ghysels and Sohn (2013), HARQ Bollerslev, Patton and Quaedvlieg (2016), DCC-NL Engle, Ledoit and Wolf (2019)
- Today: high-frequency data are cheap; the questions are statistical (noise, jumps, dimension, evaluation)

Prior knowledge from TSA and new material

- Known (TSA, Chapter 5): GARCH(1,1), GJR, EGARCH, the news impact curve, ML with Gaussian and Student-t errors, QLIKE; (TSA, Chapter 14): CCC and DCC
- New here
  - why Gaussian ML still works when returns are not Gaussian, and which standard errors are then valid
  - volatility with a slow, macro-driven component; volatility measured from intraday prices, with its own sampling error
  - models that use realised measures; evaluation against a noisy target; covariance matrices with many assets
- Stochastic volatility (a latent log-variance with its own shock): Chapter 6; Markov-switching GARCH: Chapter 7; VaR and ES backtesting: Chapter 9; rough volatility: Chapter 10

The Gaussian quasi-likelihood (1/2): the model

- The return shock is a volatility times a standardised shock; the variance follows a GARCH(1,1) recursion

$$\varepsilon_t = \sigma_t(\theta_0)\,\eta_t, \qquad \sigma_t^2(\theta) = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2(\theta)$$

- Notation
  - ε_t: the demeaned return of day t; $\sigma_t^2(\theta)$: its variance conditional on the past $\mathcal{F}_{t-1}$ (the information up to day $t-1$)
  - η_t: i.i.d. standardised shock, $\mathbb{E}\eta_t = 0$, $\mathbb{E}\eta_t^2 = 1$; its law (Normal, Student-t, skewed) is unknown
  - $\theta = (\omega, \alpha, \beta)$, true value θ_0: $\omega > 0$ sets the level, $\alpha \geq 0$ the reaction to yesterday's squared shock, $\beta \geq 0$ the persistence of yesterday's variance
- The recursion starts from an arbitrary σ_1^2; the effect of the start vanishes geometrically when $\beta < 1$

The Gaussian quasi-likelihood (2/2): estimator and score

- QMLE: maximise the Gaussian log-likelihood, used as an estimating equation even if η_t is not Gaussian

$$\hat{\theta} = \arg\max_{\theta} \sum_{t=1}^{T} \ell_t(\theta), \qquad \ell_t(\theta) = -\frac{1}{2}\Big(\ln \sigma_t^2(\theta) + \frac{\varepsilon_t^2}{\sigma_t^2(\theta)} \Big)$$

  - ℓ_t: the Gaussian log-density of day t without its constant; T: number of days; $\hat{\theta}$: the estimate
- The score, the gradient of ℓ_t, has conditional mean zero at the true value

$$s_t(\theta) = \frac{\partial \ell_t}{\partial \theta} = \frac{1}{2}\Big(\frac{\varepsilon_t^2}{\sigma_t^2} - 1 \Big) \frac{1}{\sigma_t^2} \frac{\partial \sigma_t^2}{\partial \theta}$$

  - at θ_0, $\varepsilon_t^2/\sigma_t^2 = \eta_t^2$, hence $\mathbb{E}(s_t|\mathcal{F}_{t-1}) = 0$ because $\mathbb{E}\eta_t^2 = 1$
  - only the conditional variance must be right; the shape of the law of η_t is never used
- The score is a martingale difference (conditional mean zero given the past)
  - a CLT for martingales gives asymptotic normality without independence of ε_t

Consistency and asymptotic normality (1/2): stationarity and consistency

- GARCH(1,1) is strictly stationary if and only if the log of the daily variance multiplier has a negative mean

$$\mathbb{E}\ln(\alpha_0\eta_t^2+\beta_0)<0$$

  - $\alpha_0\eta_t^2+\beta_0$: the factor by which a shock to the variance is carried from one day to the next; α_0,β_0: true values
  - by Jensen's inequality, weaker than $\alpha_0+\beta_0<1$ (finite variance): IGARCH ($\alpha_0+\beta_0=1$) is strictly stationary, and $\mathbb{E}\varepsilon_t^2$ may be infinite
  - Lee and Hansen (1994), Lumsdaine (1996): GARCH(1,1), including IGARCH; Berkes, Horváth and Kokoszka (2003), Francq and Zakoïan (2004): GARCH(p,q) under strict stationarity
- Consistency of $\hat{\theta}$ Francq and Zakoïan (2004)
  - conditions: strict stationarity, θ_0 in a compact parameter set, identifiability (η_t^2 is not a constant)
  - no moment of ε_t is needed
- Even explosive GARCH ($\mathbb{E}\ln(\alpha_0\eta^2+\beta_0)>0$): $(\hat{\alpha},\hat{\beta})$ remain consistent and asymptotically normal, ω is not identified Jensen and Rahbek (2004)

Consistency and asymptotic normality (2/2): the limit law

- With θ_0 interior to the parameter set and a finite fourth moment of η_t, the estimator is asymptotically normal

$$\sqrt{T}(\hat{\theta} - \theta_0) \to N\big(0, (\kappa_\eta - 1) J^{-1}\big), \qquad J = \mathbb{E}\Big[\frac{1}{\sigma_t^4} \frac{\partial \sigma_t^2}{\partial \theta} \frac{\partial \sigma_t^2}{\partial \theta'}\Big]$$

  - $\kappa_\eta = \mathbb{E}\eta_t^4$: kurtosis of the shock (3 for the Normal distribution); $\kappa_\eta - 1 = \mathrm{Var}(\eta_t^2)$
  - J: a 3×3 matrix, the mean outer product of the relative sensitivities $\sigma_t^{-2} \partial \sigma_t^2 / \partial \theta$; $\to$: convergence in distribution as $T \to \infty$
  - reading: the sampling variance of $\hat{\theta}$ grows linearly with the kurtosis of the shocks
- Boundary: if $\alpha_0 = 0$, θ_0 is not interior and the limit is the projection of a normal vector on a cone
  - this is why TSA, Chapter 5 tests ARCH effects one-sided

The sandwich and the Bollerslev–Wooldridge standard errors (1/2)

- General QML result: the asymptotic variance is a "sandwich" of two matrices

$$\sqrt{T}(\hat{\theta} - \theta_0) \to N(0, A^{-1} B A^{-1}), \qquad A = -\mathbb{E}\frac{\partial^2 \ell_t}{\partial\theta\,\partial\theta'}, \qquad B = \mathbb{E}\, s_t s_t'$$

  - A: the expected negative Hessian (curvature of the log-likelihood); B: the variance of the score s_t
  - if the likelihood is the true one, $A = B$ (information equality) and the variance reduces to A^{-1}; under QML, $A \neq B$

- For GARCH both matrices are proportional to J (▸ Appendix)

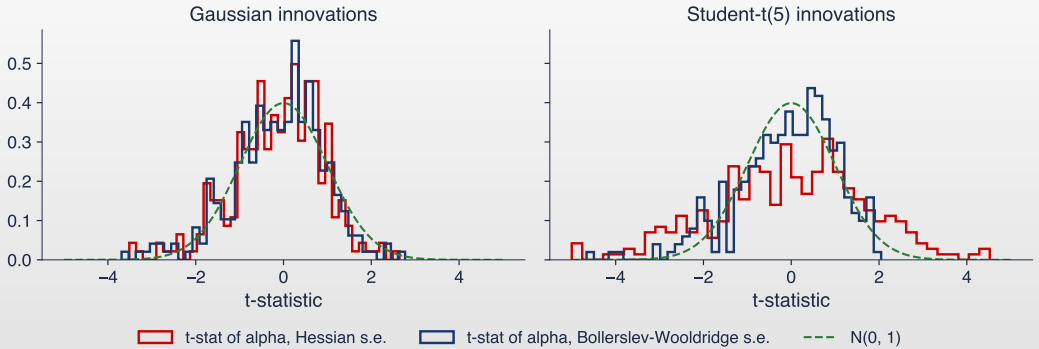
$$A = \tfrac{1}{2} J, \qquad B = \tfrac{\kappa_\eta - 1}{4} J, \qquad A^{-1} B A^{-1} = (\kappa_\eta - 1) J^{-1}$$

  - the sandwich recovers the limit law of the previous slide, for any law of η_t with $\kappa_\eta < \infty$

The sandwich and the Bollerslev–Wooldridge standard errors (2/2)

- Three ways to compute standard errors (s.e.)
  - Hessian only: $A^{-1} = 2J^{-1}$, correct only if $\kappa_\eta = 3$; too small by the factor $\sqrt{(\kappa_\eta - 1)/2}$ when $\kappa_\eta > 3$
  - outer product of the scores (OPG): $B^{-1} = \frac{4}{\kappa_\eta - 1} J^{-1}$, even smaller when $\kappa_\eta > 3$
  - sandwich Bollerslev and Wooldridge (1992): $\hat{A}^{-1}\hat{B}\hat{A}^{-1}$, valid whatever the law of η_t (with $\kappa_\eta < \infty$)
- Estimates used by Bollerslev and Wooldridge
  - $\hat{A}$: the observed Hessian of $-\frac{1}{T}\sum_t \ell_t$ at $\hat{\theta}$; $\hat{B} = \frac{1}{T}\sum_t \hat{s}_t \hat{s}_t'$, with $\hat{s}_t = s_t(\hat{\theta})$ the estimated score
- Robust Wald and score tests use the sandwich
  - the quasi-likelihood-ratio (quasi-LR) statistic is no longer χ^2, but a weighted sum of χ^2_1 variables (chi-square with one degree of freedom)

Hessian against sandwich: a Monte Carlo



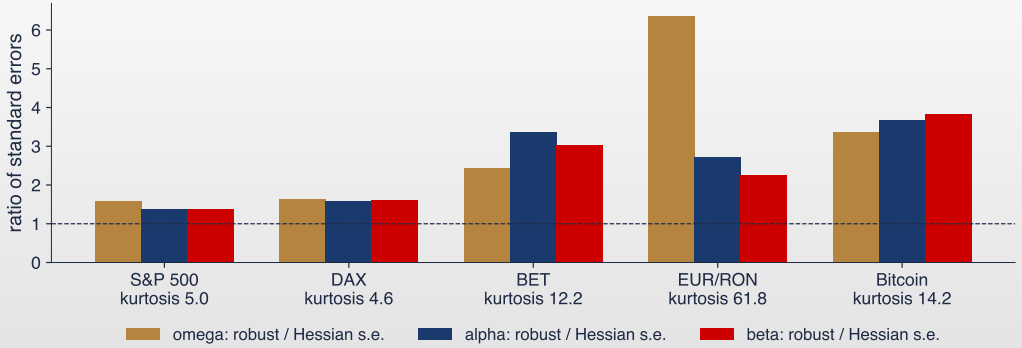
- GARCH(1,1) with $(\omega, \alpha, \beta) = (0.05, 0.08, 0.90)$, $T = 2\,000$, 300 replications; Gaussian QML in each; histogram of $(\hat{\alpha} - \alpha_0)/\widehat{\text{se}}$

ATS_ch8_qmle

Interpreting the Monte Carlo

- Gaussian η: coverage of nominal 95% intervals for α is 95.0% (Hessian) and 94.3% (sandwich): both fine
- Student-$t(5)$, $\kappa_\eta = 9$: Hessian intervals cover 73.0% for α and 74.7% for β; sandwich intervals 91.7% and 92.3%
- The theory predicts the factor $\sqrt{(9-1)/2} = 2$ between the two standard errors; the point estimates are the same
- With fat tails, a Hessian-based t-test of α rejects far too often: report sandwich standard errors by default

Robust and naive standard errors in five markets



⊡ Gaussian QML of GARCH(1,1), daily returns 2010–2026; bars: sandwich s.e. divided by Hessian s.e.; labels: kurtosis $\hat{\kappa}_\eta$ of the standardised residuals

ATS_ch8_qmle

Interpreting the five markets

- S&P 500: $\hat{\alpha} = 0.154$ with s.e. 0.013 (Hessian) against 0.018 (sandwich); $\hat{\kappa}_\eta = 5.0$, theoretical factor 1.42
- BET: $\hat{\kappa}_\eta = 12.2$, the sandwich s.e. of $\hat{\alpha}$ is 3.36 times larger; EUR/RON (managed float): 2.71 times, $\hat{\kappa}_\eta = 61.8$
- The ratios track $\sqrt{(\hat{\kappa}_\eta - 1)/2}$ only roughly: in the data η_t is not i.i.d., and the sandwich does not need it to be
- Student-t ML raises the log-likelihood by 124.4 (S&P 500) and 233.0 (BET), but is consistent only if the t shape is right; QML needs only the variance equation

GARCH with exogenous variables

- GARCH-X: yesterday's value of an observed variable enters the variance equation

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \pi' x_{t-1}$$

  - x_{t-1}: vector of exogenous variables known at the end of day $t-1$; π: their loadings ($\pi' x$: the weighted sum)
  - positivity of σ_t^2: $x_{t-1} \geq 0$ and $\pi \geq 0$, or a specification in logs
- Candidates for x_t: a realised measure (leads to HEAVY and Realized GARCH below), the VIX, a macro or policy-uncertainty index, trading volume
  - QML theory carries over if x_t is stationary and ergodic; its own dynamics need not be modelled for one-step forecasts
  - multi-step forecasts need a model for x_t: this is exactly what Realized GARCH and HEAVY add
- A slow variable (a monthly macro series) cannot enter a daily recursion at its own frequency: one needs a component structure (next slides)

The component GARCH of Engle and Lee (1/2): the model

- Engle and Lee (1999): the variance is a slowly moving long-run level q_t plus a transitory deviation from it

$$\sigma_t^2 = q_t + \alpha(\varepsilon_{t-1}^2 - q_{t-1}) + \beta(\sigma_{t-1}^2 - q_{t-1})$$

$$q_t = \omega + \rho q_{t-1} + \phi(\varepsilon_{t-1}^2 - \sigma_{t-1}^2)$$

  - q_t: long-run (permanent) component; ρ: its persistence, close to 1; ω: its intercept
  - $\sigma_t^2 - q_t$: transitory component; α, β: its reaction and persistence, with $\alpha + \beta < \rho$
  - $\varepsilon_{t-1}^2 - \sigma_{t-1}^2$: the variance surprise of yesterday (mean zero); ϕ: how much of it moves the long-run level

- Both components are driven by the same shock: the model is a restricted GARCH(2,2)

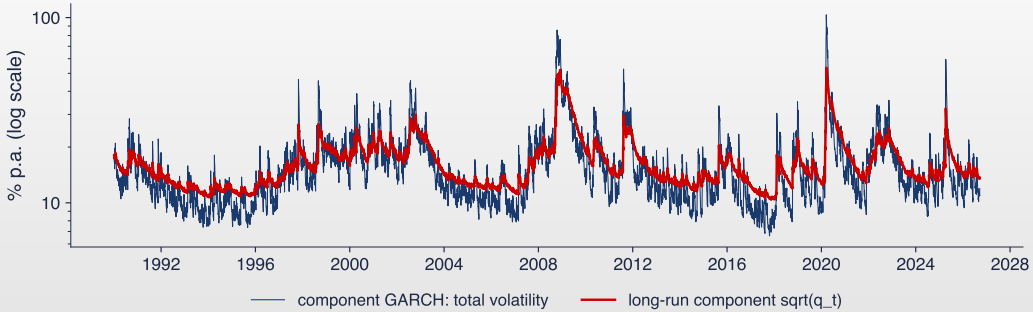
The component GARCH of Engle and Lee (2/2): the half-lives of the components

- Half-life: the number of days after which half of a shock to a component has died out

$$\mathrm{HL}_q = \frac{\ln(1/2)}{\ln \rho}, \qquad \mathrm{HL}_{\sigma-q} = \frac{\ln(1/2)}{\ln(\alpha+\beta)}$$

  - HL_q: half-life of the long-run component; $\mathrm{HL}_{\sigma-q}$: half-life of the transitory component
  - a persistence $\rho = 0.99$ gives about 69 days; $\alpha + \beta = 0.9$ gives about 6.6 days
  - GARCH(1,1) forces a single half-life, $\ln 0.5 / \ln(\alpha+\beta)$
- Two exponentials approximate a slowly decaying (hyperbolic) autocorrelation of squared returns over a finite range of lags
  - long memory proper: Chapter 10

Component GARCH for the S&P 500



- Daily returns since 1990, $T = 9\,240$; Gaussian QML; annualised total volatility $\sqrt{252\sigma_t^2}$ and long-run component $\sqrt{252q_t}$

ATS_ch8_components

Interpreting the component model

- $\hat{\rho} = 0.995$, $\hat{\phi} = 0.026$; transitory part $\hat{\alpha} + \hat{\beta} = 0.950$: half-lives 150 days (long run) and 13.6 days (short run)
- GARCH(1,1) on the same data: persistence 0.983, one half-life of 41 days, a compromise between the two
- Quasi-LR statistic against GARCH(1,1): 44.1; under the null $\phi = 0$ the parameter ρ is not identified (Davies problem), so the χ^2_2 reference is only indicative
- After 2008, 2020 and 2025 the long-run level stays high for months while daily volatility has already fallen: this matters for horizons beyond a few weeks

GARCH-MIDAS (1/3): a macro-driven long-run component



Eric Ghysels, 2019

Photo: Eghysels (2019); CC BY-SA 4.0; Wikimedia Commons

- Engle, Ghysels and Sohn (2013): the daily variance is the product of a monthly level τ_t and a daily factor $g_{i,t}$

$$r_{i,t} = \mu + \sqrt{\tau_t \, g_{i,t}} \, \eta_{i,t}$$

  - $r_{i,t}$: return of day i of month t; μ: mean return; $\eta_{i,t}$: i.i.d. shock with mean 0 and variance 1

- Short run: a GARCH(1,1) with mean 1, re-scaled by τ_t

$$g_{i,t} = (1 - \alpha - \beta) + \alpha \frac{(r_{i-\mathbf{1},t} - \mu)^{\mathbf{2}}}{\tau_t} + \beta g_{i-\mathbf{1},t}$$

  - α, β: reaction and persistence of the daily component, as in GARCH(1,1)

GARCH-MIDAS (2/3): the long-run component

- Long run: MIDAS (mixed-data sampling) regression of $\ln \tau_t$ on K lags of a monthly variable

$$\ln \tau_t = m + \theta \sum_{k=1}^{K} \varphi_k(w)\, X_{t-k}, \qquad \varphi_k(w) \propto (1 - k/K)^{w-1}$$

  - X_{t-k}: the monthly driver k months ago; m: intercept; θ: effect of the driver (sign: pro- or countercyclical)
  - $\varphi_k(w)$: beta lag weights, normalised to sum to 1 and restricted to decay Ghysels, Santa-Clara and Valkanov (2006); $w \geq 1$: the larger w, the faster the decay; $w = 1$: equal weights
- Choices of X
  - realised variance of past months, in levels: $\tau_t = m + \theta \sum_k \varphi_k(w) \mathrm{RV}_{t-k}$, with RV_{t-k} the sum of squared daily returns of month $t - k$
  - macro data: industrial production growth, PPI inflation

GARCH-MIDAS (3/3): identification, estimation and the variance ratio

- τ_t is constant within the month and $\mathbb{E} g_{i,t} = 1$: the split is identified by the different frequencies, without a filter for a latent variable
  - Gaussian QML of all parameters $(\mu, \alpha, \beta, m, \theta, w)$ in one step; sandwich standard errors
- Variance ratio: the share of the variation of log volatility explained by the long-run component

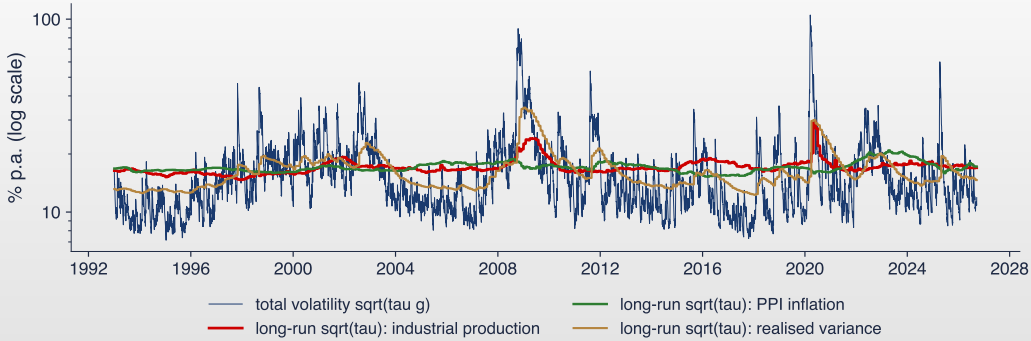
$$\mathrm{VR} = \frac{\mathrm{Var}(\ln \tau_t)}{\mathrm{Var}(\ln(\tau_t \, g_{i,t}))} \in [0, 1]$$

  - VR close to 0: the long-run component is almost flat; close to 1: it carries most of the variation
  - a macro variable can be significant ($\hat{\theta} \neq 0$) and still explain little of daily volatility (small VR)
- Inference on w is nonstandard when $\theta = 0$ (w is then not identified) and at the boundary $w = 1$
- Extensions: two-sided weights, several regressors, a second (daily) component Conrad and Kleen (2020); forecasts beyond one month use only the long-run part

Case study: Engle, Ghysels and Sohn (2013) on our data

- **Question** of Engle, Ghysels and Sohn (2013): does the stock market's long-run volatility move with macroeconomic fundamentals (output growth, inflation)?
  - their sample of US stock returns is much longer than ours; their benchmark long-run component is a fixed-window realised variance
- **Our replication**: S&P 500 daily returns, common sample from 04.01.1993, $T = 8\,482$; $K = 36$ monthly lags; restricted beta weights
  - three long-run drivers: monthly realised variance (sum of squared daily returns), industrial production growth (FRED INDPRO), PPI inflation (FRED WPSFD49207)
  - all models against GARCH(1,1) on the same days; QML with sandwich standard errors
- Extension for a project: Romanian industrial production and the BET (Seminar 8, B5); real-time vintages of the macro data

GARCH-MIDAS for the S&P 500



- Annualised $\sqrt{252\tau_t g_{i,t}}$ of the industrial-production model and the long-run components $\sqrt{252\tau_t}$ of the three models

ATS_ch8_components

GARCH-MIDAS estimates

Long-run driver	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\theta}$ (s.e.)	$\hat{w}$	VR (%)	BIC
realised variance (levels)	0.119	0.841	0.0225 (0.0058)	5.939	25.1	22710.0
industrial production growth	0.110	0.870	−0.551 (0.353)	3.692	4.5	22732.5
PPI inflation	0.108	0.873	0.753 (1.260)	1.000	2.2	22733.3
GARCH(1,1)						22721.9

- Same days for all models; BIC $= -2 \ln L + k \ln T$ (lower is better); s.e.: Bollerslev–Wooldridge

Interpreting GARCH-MIDAS

- Realised variance as the slow driver: $\hat{\theta} = 0.0225$ ($t = 3.86$), VR = 25.1%, BIC below GARCH(1,1): a long-run component exists
- Industrial production: $\hat{\theta} = -0.551$ ($t = -1.56$), the countercyclical sign of Engle, Ghysels and Sohn (2013), but VR only 4.5% and a higher BIC than GARCH(1,1)
- PPI inflation: $\hat{w}$ at the boundary 1 (flat weights) and $\hat{\theta}$ imprecise: no evidence in 1993–2026
- Replication verdict: the qualitative sign survives, the strength does not; macro effects need long samples with deep recessions, which is why the original study uses a much longer history

Prices as Itô semimartingales (1/2): the model

- Within day t, the log price moves by a drift, a Brownian part scaled by the spot volatility, and jumps

$$dX_s = \mu_s\, ds + \sigma_s\, dW_s + dJ_s, \qquad s \in [0,1]$$

  - X_s: log price at intraday time s (the day is rescaled to $[0,1]$); μ_s: drift; W_s: standard Brownian motion
  - σ_s: stochastic spot (instantaneous) volatility; J_s: a jump process with finitely many jumps per day, ΔJ_s: the jump at time s

- Quadratic variation of the day: the diffusive part plus the squared jumps

$$\mathrm{QV}_t = \mathrm{IV}_t + \sum_{s \le 1} (\Delta J_s)^2, \qquad \mathrm{IV}_t = \int_0^1 \sigma_s^2\, ds$$

  - IV_t: integrated variance, the average of the spot variance over the day
  - without jumps and with σ independent of W: $r_t | \mathrm{IV}_t \sim N(\int_0^1 \mu_s\, ds, \mathrm{IV}_t)$, so IV is the variance that matters for the daily return r_t

Prices as Itô semimartingales (2/2): realised variance

- Realised variance: the sum of the n squared intraday returns of the day

$$\mathrm{RV}_t = \sum_{i=1}^{n} r_{t,i}^2, \qquad r_{t,i} = X_{i/n} - X_{(i-1)/n}$$

  - n: number of intraday intervals (78 five-minute returns in a 6.5-hour session); $r_{t,i}$: log return over the i-th interval
  - $\mathrm{RV}_t \to \mathrm{QV}_t$ in probability as $n \to \infty$: RV measures the total (diffusive plus jump) variation
- The drift does not matter in the limit: it is $O(1/n)$ per interval, while the Brownian part is $O(n^{-1/2})$
  - volatility becomes observable *ex post*, without a model Andersen, Bollerslev, Diebold and Labys (2003), Barndorff-Nielsen and Shephard (2002)

The central limit theorem for realised variance (1/2)

- Barndorff-Nielsen and Shephard (2002): without jumps, the error of RV shrinks at rate $\sqrt{n}$ and is mixed normal

$$\sqrt{n}\,(\mathrm{RV}_t - \mathrm{IV}_t) \to MN(0, 2\,\mathrm{IQ}_t), \qquad \mathrm{IQ}_t = \int_0^1 \sigma_s^4\, ds$$

  - IQ_t: integrated quarticity, the daily average of σ_s^4; it sets the size of the error
  - *MN* (mixed normal): normal given the path of σ, with a random variance $2\,\mathrm{IQ}_t$
  - the convergence is *stable* in law: one may divide by a random, consistently estimated $\sqrt{\mathrm{IQ}_t}$ and keep the limit

- Rate $\sqrt{n}$: with $n = 78$ and volatility constant within the day, the relative error is about $\sqrt{2/78} \approx 16\%$
  - the measurement error $\mathrm{RV}_t - \mathrm{IV}_t$ is heteroskedastic (it scales with IQ_t): HARQ, below, is built on this fact

The central limit theorem for realised variance (2/2): feasible intervals

- Feasible version: estimate IQ by the realised quarticity RQ, then studentise

$$\widehat{\mathrm{IQ}}_t = \mathrm{RQ}_t = \frac{n}{3}\sum_{i=1}^{n} r_{t,i}^4, \qquad \frac{\mathrm{RV}_t - \mathrm{IV}_t}{\sqrt{\frac{2}{3}\sum_i r_{t,i}^4}} \to N(0,1)$$

  - the factor $\frac{n}{3}$ makes RQ unbiased under constant volatility, because $\mathbb{E}Z^4 = 3$ for $Z \sim N(0,1)$
  - 95% interval for IV_t: $\mathrm{RV}_t \pm 1.96\sqrt{\frac{2}{3}\sum_i r_{t,i}^4}$; it may contain negative values

- Log version (delta method): an interval for $\ln \mathrm{IV}_t$

$$\ln \mathrm{RV}_t \pm 1.96\,\frac{\sqrt{\frac{2}{3}\sum_i r_{t,i}^4}}{\mathrm{RV}_t}$$

  - exponentiated, it never contains negative values and has better coverage in finite samples

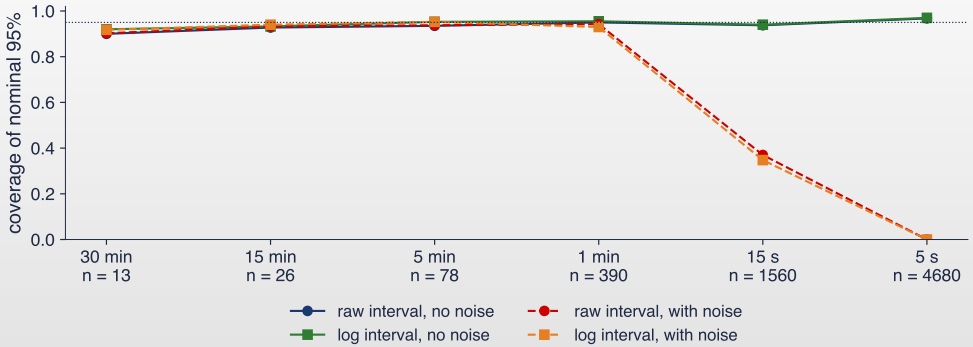
Simulated prices with known integrated variance

- One-second grid, $n = 23\,400$ seconds a day; the log spot variance is the sum of two Gaussian AR(1) factors

$$\ln \sigma_s^2 = c + f_s^{(1)} + f_s^{(2)}$$

  - c: constant fixing the mean; $f^{(1)}$: slow factor (half-life 60 days, stationary s.d. 0.75); $f^{(2)}$: fast factor (half-life 2 days, s.d. 0.45)
  - mean IV = 1 (%^{2} a day), the mean of the S&P 500 realised kernel in 2000–2022; leverage: correlation -0.6 between price shocks and shocks to the fast factor
- Jumps: Poisson with 0.08 jumps a day, sizes $N(0, 0.8^2)$ (%): jumps make about 5.6% of quadratic variation
- Noise: we observe $Y_s = X_s + u_s$, with u_s i.i.d. $N(0, \omega^2)$ and $\omega = 0.004\%$
  - Y_s: observed log price; X_s: efficient log price; u_s: microstructure noise with standard deviation ω; at one second, $2n\omega^2 = 0.75$ (slide Microstructure noise)
- Each estimator below is compared with the true IV or QV of the same simulated day

Coverage of the feasible CLT



- 400 simulated days without jumps; 95% intervals for IV_t (raw and log) from returns sampled every 30 minutes to every 5 seconds, without and with noise

ATS_ch8_realised_measures

Interpreting the CLT coverage

- Without noise, coverage approaches 95% as n grows: raw interval 90.0% at $n = 13$ and 95.0% at $n = 390$; the log interval is closer at small n (92.0%)
- With noise, coverage collapses once sampling is too fine: 37.0% at 15 seconds and 0.0% at 5 seconds; at 1 minute it is still 94.2%
- The interval shrinks like $n^{-1/2}$ while the noise bias grows like n: the CLT is a statement about the efficient price, not about the observed one
- Practical rule: 5-minute returns for liquid assets unless a noise-robust estimator is used

Where the realised measures come from

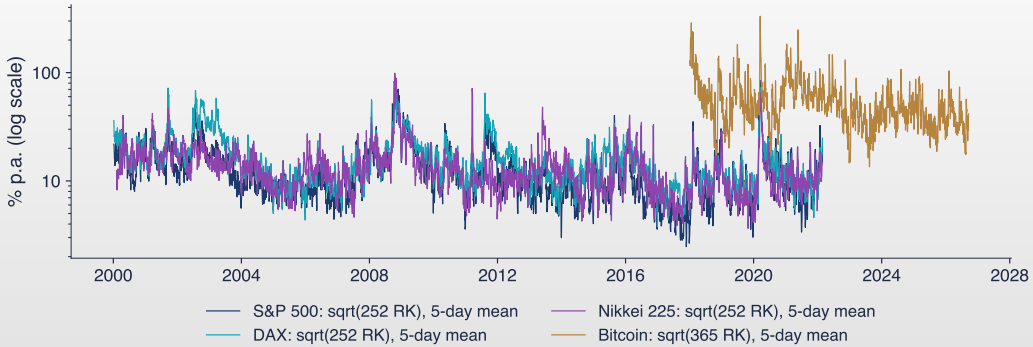


Ole E. Barndorff-Nielsen, 2007

Photo: Thomas Steiner (2007); CC BY-SA 2.5; Wikimedia Commons

- Heber, Lunde, Shephard and Sheppard (2009), version 0.3: daily RV, BV, realised kernels and open-to-close returns for 31 indices, 2000 to 25 February 2022, built from cleaned tick data
- its terms: free use if the library and its version are cited; the archived copy keeps the 2022 file available after the shutdown
- Bitcoin and Ether trade 24/7 and Binance publishes all one-minute and one-second prices: we compute RV, BV, RQ, tripower quarticity and a realised kernel for each UTC day, 2018–2026
- No free source covers equity intraday data after 2022: a project can rebuild them from a licensed feed

Realised volatility across markets



⊡ Annualised realised kernel, 5-day means: S&P 500, DAX and Nikkei 225 (Oxford-Man, 2000–2022); Bitcoin (Binance one-minute prices, 2018–2026, 365 days a year)

ATS_ch8_realised_measures

Interpreting the realised volatilities

- Average annualised volatility from the kernel: S&P 500 16.2%, DAX 19.5%, Nikkei 16.6%, Bitcoin 68.4%
- The equity series move together (2002, 2008, 2011, 2020): a common long-run factor, the multivariate theme of the last section
- Open-to-close measures miss the overnight return: for daily risk one adds the squared overnight return or scales the measure
- Bitcoin has no overnight gap, but its daily volatility is three to four times higher and falls after 2023

Microstructure noise (1/2): the bias of RV

- The observed log price is the efficient price plus a noise term

$$Y_{i/n} = X_{i/n} + u_i$$

  - u_i: i.i.d. noise with variance ω^2, independent of X; sources: bid–ask bounce, price discreteness, stale quotes

- RV computed from the observed prices at n intervals, $\mathrm{RV}_t^{(n)}$, is biased upwards, and the bias grows with n

$$\mathbb{E}(\mathrm{RV}_t^{(n)}|X) = \mathrm{IV}_t + 2n\omega^2, \qquad \mathrm{Var}(\mathrm{RV}_t^{(n)}|X) \approx 4n\,\mathbb{E}u^4$$

  - each observed return contains $u_i - u_{i-1}$, with variance $2\omega^2$; summed over n returns this gives $2n\omega^2$
  - RV diverges as $n \to \infty$: sampling as often as possible is not optimal
  - derivation of the bias and of the variance: Appendix

Microstructure noise (2/2): diagnostics

- Two consequences of i.i.d. noise
  - $\hat{\omega}^2 = \mathrm{RV}_t^{(n)}/(2n)$ at the highest frequency estimates the noise variance, since $2n\omega^2$ dominates IV there
  - returns of the observed price have first-order autocovariance $-\omega^2$: negative autocorrelation is the signature of i.i.d. noise
- Signature plot: average $\mathrm{RV}^{(n)}$ against the sampling interval
  - a flat region shows the intervals where noise no longer matters
- Real noise is not i.i.d.: it is autocorrelated and correlated with the efficient price, especially in quote data Hansen and Lunde (2006b)

Optimal sparse sampling

- Mean square error (MSE) of $\mathrm{RV}^{(n)}$ as an estimator of IV: sampling variance falls, noise bias rises with n

$$\mathrm{MSE}(n) \approx \underbrace{\frac{2\,\mathrm{IQ}}{n}}_{\text{variance}} + \underbrace{(2n\omega^2)^2}_{\text{bias}^2} + 4n\,\mathbb{E}u^4 + \dots$$

- Minimising the first two terms gives the optimal number of intervals Bandi and Russell (2008)

$$n^* \approx \left(\frac{\mathrm{IQ}}{4\omega^4}\right)^{1/3}$$

  - the optimal grid is coarser when noise (ω^2) is large relative to volatility (IQ)
  - example (Seminar 8, A3): IV = IQ = 1, $\omega^2 = 1.6 \times 10^{-5}$ gives $n^* \approx 992$, a return every 23.6 seconds

- The best sparse RV converges only at rate $n^{1/6}$ and discards most data: this motivates estimators that use all observations
- Subsampling: average the RV of the K offset grids (starting at second $0, 1, \dots, K-1$); same bias, smaller variance

Two scales and realised kernels (1/2)

- Two-scales RV Zhang, Mykland and Aït-Sahalia (2005): the fast scale estimates the noise bias of the slow one and removes it

$$\mathrm{TSRV} = \overline{\mathrm{RV}}^{(K)} - \frac{\bar{n}}{n}\,\mathrm{RV}^{(n)}, \qquad \bar{n} = \frac{n - K + 1}{K}$$

  - $\overline{\mathrm{RV}}^{(K)}$: subsampled RV on K offset sparse grids (slow scale), each with about $\bar{n}$ returns; $\mathrm{RV}^{(n)}$: RV on all n returns (fast scale)
  - the bias of the slow scale is $2\bar{n}\omega^2 = \frac{\bar{n}}{n} \cdot 2n\omega^2$: the correction subtracts it; rate $n^{1/6}$
- Pre-averaging Jacod et al. (2009): average returns over blocks before squaring
  - rate $n^{1/4}$, the optimal one in the presence of noise; in practice similar to the realised kernel

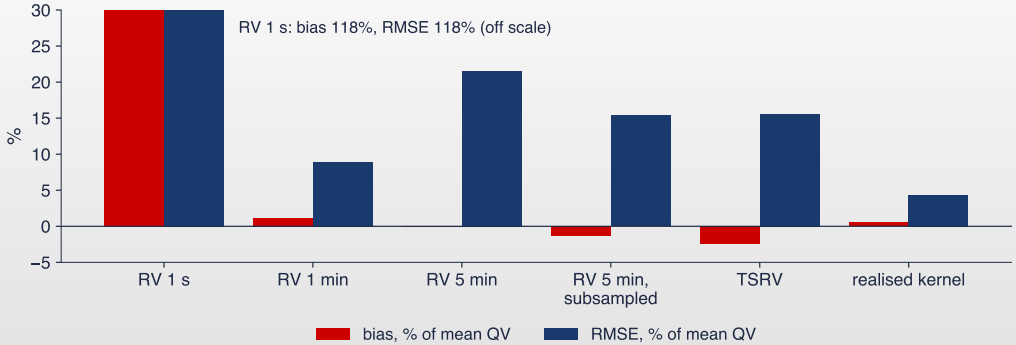
Two scales and realised kernels (2/2): the realised kernel

- Realised kernel Barndorff-Nielsen, Hansen, Lunde and Shephard (2008): RV plus weighted realised autocovariances

$$\mathrm{RK}_t = \gamma_0 + \sum_{h=1}^{H} k\Big(\frac{h}{H+1}\Big)(\gamma_h + \gamma_{-h}), \qquad \gamma_h = \sum_i r_{t,i}\, r_{t,i-h}$$

  - γ_h: realised autocovariance at lag h ($\gamma_0 = \mathrm{RV}$); H: bandwidth, the number of lags used; $k(\cdot)$: weight function with $k(0) = 1$, $k(1) = 0$
  - Parzen weights: $k(x) = 1 - 6x^2 + 6x^3$ for $x \le \frac{1}{2}$ and $2(1-x)^3$ for $\frac{1}{2} < x \le 1$
  - the negative γ_1 created by noise cancels the bias $2n\omega^2$: the same idea as HAC variance estimation (Chapter 0)
- Properties
  - nonnegative by construction; rate $n^{1/5}$ with $H = c^*\xi^{4/5}n^{3/5}$, where $\xi^2 = \omega^2/\sqrt{\mathrm{IQ}}$ is the noise-to-signal ratio and $c^* = 3.51$ for Parzen Barndorff-Nielsen et al. (2009)
  - it also corrects serially dependent noise of short range, of either sign

Estimators against the true quadratic variation



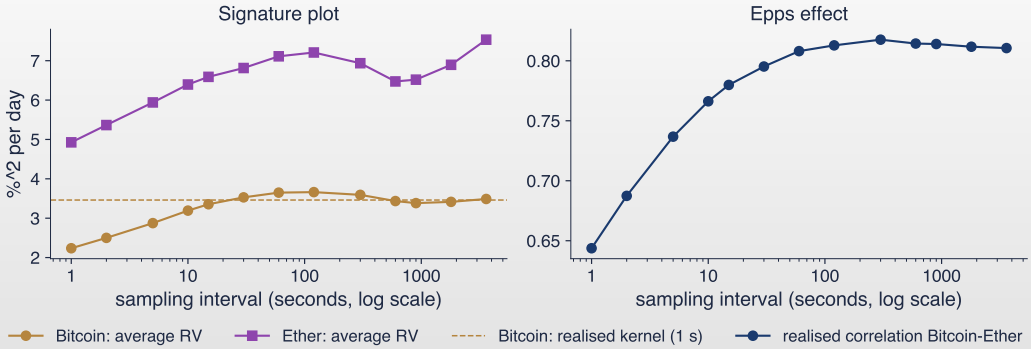
- 300 simulated days with jumps and noise; bias and root mean square error relative to the mean QV; TSRV with $K = 300$ seconds; RK with the BNHLS bandwidth (median $H = 31$)

ATS_ch8_realised_measures

Interpreting the estimator comparison

- RV at one second is dominated by noise: bias 117.8% of QV; at one minute the bias is 1.2% and the RMSE 8.9%
- Five-minute RV is unbiased but noisy (RMSE 21.6%); subsampling and TSRV reduce it to 15.4% and 15.6%
- The realised kernel uses all 23 400 returns: bias 0.5%, RMSE 4.4%, the best of the six
- All estimators here target QV, jumps included; separating the jump part needs the next section

Signature plot and Epps effect: Bitcoin and Ether



- Binance one-second prices, August 2026 (31 days); average daily RV by sampling interval (subsampled); realised correlation of the two coins by interval

ATS_ch8_realised_measures

Interpreting the signature plot

- Bitcoin: RV is 2.24 at one second, 3.65 at one minute, 3.59 at five minutes: the signature *rises*, the opposite of the i.i.d.-noise prediction
- Reason: 52% of one-second returns are zero (no trade or the same price) and prices adjust gradually, so one-second returns are positively autocorrelated
- The realised kernel on one-second data (3.46, median $H = 22$) adds the positive autocovariances back and lands on the flat region of 1–30 minutes
- Epps effect Epps (1979): realised correlation 0.64 at one second against 0.82 at five minutes; asynchronous trading biases covariances towards zero at high frequency

Bipower variation (1/2)

- Barndorff-Nielsen and Shephard (2004): sum of products of adjacent absolute returns, rescaled

$$\mathrm{BV}_t = \mu_1^{-2} \frac{n}{n-1} \sum_{i=2}^{n} |r_{t,i}|\,|r_{t,i-1}|, \qquad \mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi}$$

  - $Z \sim N(0,1)$; μ_1^{-2} makes BV_t consistent for IV_t; $\frac{n}{n-1}$ corrects for the $n-1$ products
  - $\mathrm{BV}_t \to \mathrm{IV}_t$ even with (finite-activity) jumps: a jump enters only two products, each multiplied by a return of order $n^{-1/2}$
- Hence $\mathrm{RV}_t - \mathrm{BV}_t \to \sum_{s \le 1} (\Delta J_s)^2$: a nonparametric estimate of the jump variation of the day

Bipower variation (2/2): related estimators

- Jump-robust quarticity: the tripower quarticity

$$\mathrm{TQ}_t = n\,\mu_{4/3}^{-3}\,\frac{n}{n-2}\sum_{i=3}^{n}\prod_{j=0}^{2}|r_{t,i-j}|^{4/3}, \qquad \mu_{4/3} = \mathbb{E}|Z|^{4/3}$$

  - products of three adjacent absolute returns, each to the power 4/3; $\mathrm{TQ}_t \to \mathrm{IQ}_t$ even with jumps (RQ does not)
- Nearest-neighbour truncation (MedRV, MinRV) Andersen, Dobrev and Schaumburg (2012): smaller finite-sample bias from a jump and from zero returns
- Threshold (truncated) RV: drop the returns above $c\,n^{-\varpi}$
  - $c > 0$: a constant (a multiple of the local volatility); $\varpi \in (0, \frac{1}{2})$: the threshold shrinks more slowly than a diffusive return, so only jumps are removed

Testing for jumps (1/2): the ratio statistic

- Barndorff-Nielsen and Shephard (2006): without jumps, the difference RV − BV is also mixed normal

$$\sqrt{n}(\mathrm{RV}_t - \mathrm{BV}_t) \to MN(0, \vartheta\, \mathrm{IQ}_t), \qquad \vartheta = \mu_1^{-4} + 2\mu_1^{-2} - 5 = \frac{\pi^2}{4} + \pi - 5 \approx 0.609$$

  - ϑ: a constant that measures how much less efficient BV is than RV
- Ratio statistic Huang and Tauchen (2005): the relative jump contribution, studentised

$$z_t = \frac{(\mathrm{RV}_t - \mathrm{BV}_t)/\mathrm{RV}_t}{\sqrt{\frac{\vartheta}{n} \max\left(1, \mathrm{TQ}_t/\mathrm{BV}_t^2\right)}} \to N(0,1)$$

  - one-sided test: reject "no jump on day t" for large z_t
  - the ratio form and the max adjustment give the best size in their simulations; $\mathrm{TQ}_t/\mathrm{BV}_t^2 \geq 1$ by Jensen when volatility varies within the day

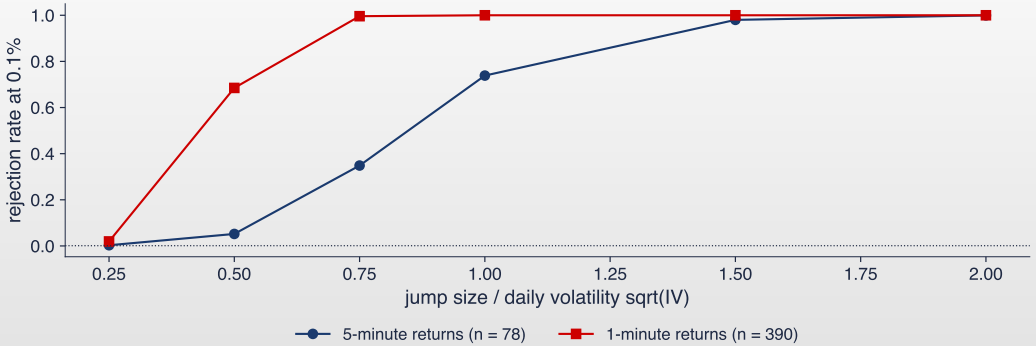
Testing for jumps (2/2): in practice

- A daily test answers "was there a jump today?"; locating it within the day needs a test on each return standardised by local BV Lee and Mykland (2008)
- Multiple testing: at level α over T days one expects αT false jump days
  - use a small α (0.1%) or a family-wise (FWER) or false-discovery-rate (FDR) correction (Chapter 1)
- Separating the continuous and jump parts of RV Andersen, Bollerslev and Diebold (2007)

$$J_t = \mathbb{1}(z_t > z_{1-\alpha})\,(\mathrm{RV}_t - \mathrm{BV}_t), \qquad C_t = \mathrm{RV}_t - J_t$$

  - $\mathbb{1}(\cdot)$: indicator, 1 if the test rejects and 0 otherwise; $z_{1-\alpha}$: the $1-\alpha$ quantile of $N(0,1)$
  - J_t: jump part, nonzero only on significant days; C_t: continuous part

Size and power of the ratio test



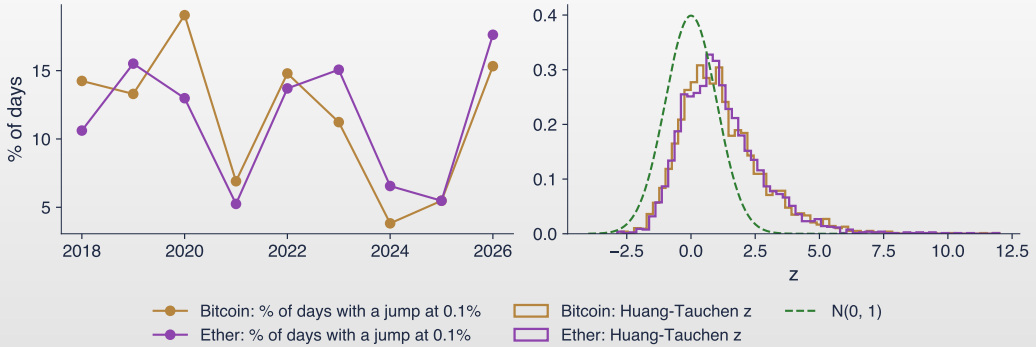
- 2 000 simulated days (two-factor SV, no jumps), then one jump of size $c\sqrt{\mathrm{IV}_t}$ at a random time; one-sided test at 0.1% with 5-minute and 1-minute returns

ATS_ch8_jumps

Interpreting the size and power

- Size without jumps: 0.0% (5 minutes) and 0.2% (1 minute) against the nominal 0.1%; with i.i.d. noise of the calibrated size still 0.1% at 1 minute
- Power for a jump of half a daily standard deviation: 5% with 5-minute returns, 68% with 1-minute returns; for one standard deviation 74% and 100%
- The power depends on $c\sqrt{n}$: small jumps are absorbed into the diffusion at coarse sampling (Seminar 8, A6 gives the detection threshold)
- The trade-off with noise is the same as for RV: finer sampling raises power until noise and zero returns distort the statistic

Jump days of Bitcoin and Ether



- Huang–Tauchen test with 5-minute returns ($n = 288$ a UTC day), 2018–2026, $T = 3\,165$ days; left: share of jump days at 0.1% by year; right: distribution of z_t against $N(0, 1)$

ATS_ch8_jumps

Interpreting the crypto jump tests

- Bitcoin: 362 jump days (11.4%) where about 3.2 false rejections are expected; Ether: 355 days (11.2%)
- Yet the jumps are small: they make 3.0% (Bitcoin) and 2.3% (Ether) of total variation; the rest is continuous
- The whole distribution of z_t is shifted (mean 1.16, not 0): besides jumps, 24-hour intraday seasonality, zero returns and flash moves violate the assumptions of the null
- Before calling a day a jump day: standardise returns by an intraday volatility pattern, check zero returns, and control the number of tests

The HAR model of Corsi (1/2): the regression

- Corsi (2009): tomorrow's RV is a linear function of the daily, weekly and monthly averages of past RV

$$\mathrm{RV}_{t+1} = \beta_0 + \beta_d \mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + u_{t+1}$$

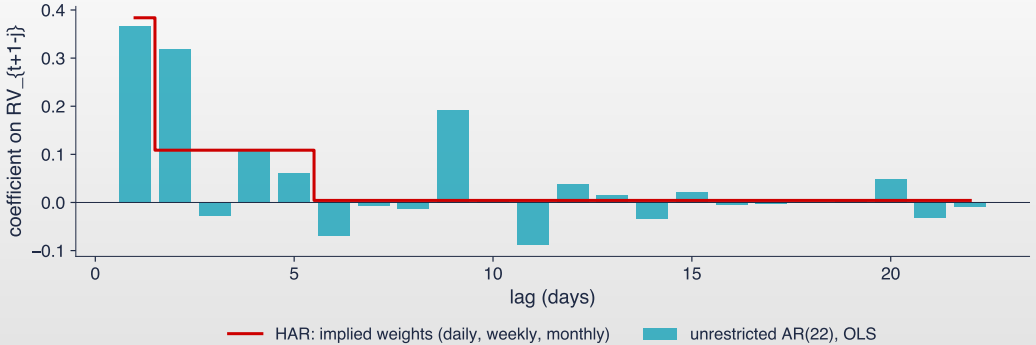
$$\mathrm{RV}_t^{(w)} = \frac{1}{5}\sum_{j=0}^{4} \mathrm{RV}_{t-j}, \qquad \mathrm{RV}_t^{(m)} = \frac{1}{22}\sum_{j=0}^{21} \mathrm{RV}_{t-j}$$

  - $\mathrm{RV}_t^{(w)}$, $\mathrm{RV}_t^{(m)}$: averages over the last 5 trading days (a week) and the last 22 (a month)
  - $\beta_d, \beta_w, \beta_m \geq 0$: weights of the three horizons; β_0: intercept; u_{t+1}: forecast error with mean zero
  - $\beta_d + \beta_w + \beta_m$: the persistence; the closer to 1, the slower volatility returns to its mean
- Heterogeneous market hypothesis: daily, weekly and monthly traders react to volatility at their own horizon
  - the cascade produces slowly decaying autocorrelations

The HAR model of Corsi (2/2): estimation and variants

- Statistically, HAR is an AR(22) with 19 linear restrictions (step-shaped coefficients)
  - short memory, but it mimics long memory over a month
- OLS is consistent; the errors are heteroskedastic and, for multi-day targets, overlapping
  - use Newey–West Newey and West (1987) (HAC) standard errors
- Variants
  - log-HAR: the same regression on $\ln \mathrm{RV}$, with near-Gaussian errors; the forecast of RV is $\exp(\hat{\mu} + \frac{1}{2}\hat{\sigma}_u^2)$, with $\hat{\mu}$ the fitted log value and $\hat{\sigma}_u^2$ the residual variance
  - HAR on $\sqrt{\mathrm{RV}}$; h-day targets (the average RV over the next h days)
- Extensions: continuous and jump parts, HAR-CJ Andersen, Bollerslev and Diebold (2007); positive and negative semivariances Patton and Sheppard (2015); implied volatility; leverage terms

HAR as a restricted AR(22)



- S&P 500 (Oxford-Man, 5-minute RV, 2000–2022, $T = 5\,530$): coefficients of an unrestricted AR(22) by OLS and the step weights implied by the HAR estimates

ATS_ch8_har

Interpreting the HAR estimates

- HAR: $\hat{\beta}_d = 0.275$ (0.104), $\hat{\beta}_w = 0.522$ (0.140), $\hat{\beta}_m = 0.094$ (0.078), Newey–West s.e.; $R^2 = 0.560$; persistence $\sum \hat{\beta} = 0.891$
- The AR(22) with 23 coefficients reaches $R^2 = 0.596$ in sample, with erratic coefficients; HAR captures the shape with 4
- Log-HAR: $\hat{\beta}_d = 0.393$, $\hat{\beta}_w = 0.399$, $\hat{\beta}_m = 0.153$, $R^2 = 0.719$ on $\ln \mathrm{RV}$; in logs the monthly component matters more
- In levels a few crisis days dominate OLS: large standard errors on β_d and β_m; weighted least squares or logs give more stable estimates

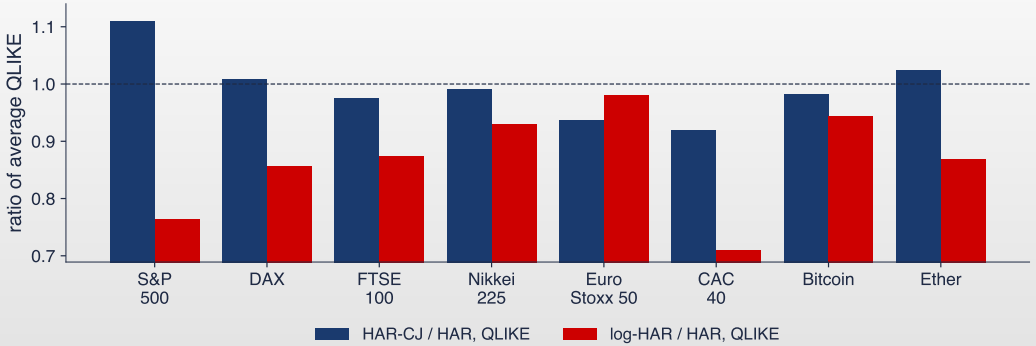
Forecasting design

- One day ahead, rolling window of 1 000 days, re-estimated every day; target RV_{t+1}
- Losses: MSE $= (\mathrm{RV} - f)^2$ and QLIKE, both robust to the noise in RV (see below)

$$\mathrm{QLIKE} = \frac{\mathrm{RV}}{f} - \ln \frac{\mathrm{RV}}{f} - 1 \geq 0$$

  - f: the variance forecast; QLIKE is 0 when $f = \mathrm{RV}$ and penalises under-prediction more than over-prediction
- Insanity filter Bollerslev, Patton and Quaedvlieg (2016): a forecast outside the range of the in-sample RV is replaced by the in-sample mean
  - linear HAR-type forecasts can be negative after a spike; QLIKE is then undefined
  - the filter is part of the method and must be reported
- Diebold–Mariano Diebold and Mariano (1995) t-statistic of the loss differential against HAR
  - Newey–West variance; negative = better than HAR, below -1.96 significant at 5%; several models: the model confidence set Hansen, Lunde and Nason (2011), Chapter 1
- Models: HAR, HAR-CJ (continuous part and jump part, from BV), log-HAR; for Bitcoin and Ether also HARQ

HAR extensions out of sample



- Average QLIKE relative to HAR (below 1 = better), six Oxford-Man indices (2004–2022) and two cryptocurrencies (from 01.11.2020); 1 000-day rolling window

ATS_ch8_har

Interpreting the out-of-sample HAR comparison

- Log-HAR beats HAR in QLIKE for 6 of 6 indices: S&P 500 0.764 (DM -2.07), CAC 40 0.709 (DM -1.73), FTSE 0.874 (DM -5.08)
- HAR-CJ helps in 4 of 6 indices, but not for the S&P 500 (1.110): separating jumps adds estimation noise when jumps are small
- The ranking depends on the loss: in MSE the gains of log-HAR are smaller (S&P 500: 0.837); MSE weighs the crisis days most
- Six indices from one data vendor and one period: report all of them, not the best one (data snooping, Chapter 1)

HARQ (1/2): the measurement error of RV

- RV is the latent IV plus a measurement error whose variance changes every day

$$\mathrm{RV}_t = \mathrm{IV}_t + e_t, \qquad \mathrm{Var}(e_t|\mathrm{IQ}_t) \approx \frac{2\,\mathrm{IQ}_t}{n}$$

  - e_t: measurement error of day t; on volatile days (IQ_t large) RV is less precise

- Errors in variables: in an AR(1) for IV with coefficient ϕ, the OLS slope on RV is attenuated (Appendix)

$$\operatorname{plim} \hat{\phi} = \phi\lambda, \qquad \lambda = \frac{\mathrm{Var}(\mathrm{IV})}{\mathrm{Var}(\mathrm{IV}) + \mathbb{E}\mathrm{Var}(e)} \in (0,1)$$

  - λ: reliability ratio, the share of the variance of RV that is signal
  - one constant β_d is a compromise: too high on noisy days, too low on precise days

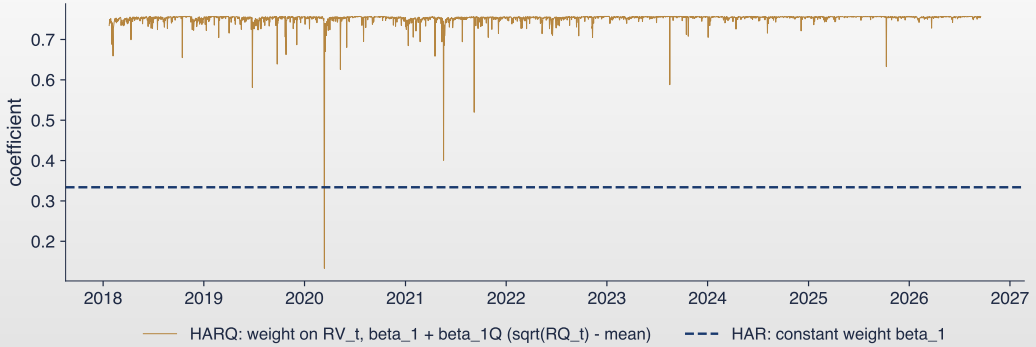
HARQ (2/2): a weight that depends on precision

- Bollerslev, Patton and Quaedvlieg (2016): the weight of yesterday's RV moves with its estimated precision

$$\mathrm{RV}_{t+1} = \beta_0 + \left(\beta_d + \beta_{dQ}\sqrt{\mathrm{RQ}_t}\right)\mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + u_{t+1}$$

  - RQ_t: realised quarticity, the estimate of IQ_t; $\sqrt{\mathrm{RQ}_t}$ is proportional to the standard deviation of e_t
  - $\beta_{dQ} < 0$ expected: the less precise yesterday's RV, the lower its weight
  - $\sqrt{\mathrm{RQ}_t}$ is demeaned, so β_d is the weight on a day of average precision
- One extra parameter, still OLS
- The original paper: S&P 500 futures and 27 Dow Jones stocks, gains over HAR in and out of sample
  - our replication extends it to Bitcoin and Ether, 2018–2026, where RQ is available from our one-minute data

HARQ for Bitcoin: a weight that moves with precision



- Full-sample HARQ on Bitcoin 5-minute RV and RQ: daily weight $\hat{\beta}_d + \hat{\beta}_{dQ}(\sqrt{\mathrm{RQ}_t} - \overline{\sqrt{\mathrm{RQ}}})$ against the constant HAR weight

ATS_ch8_har

Interpreting HARQ

- $\hat{\beta}_{dQ} = -0.00021$ ($t = -2.7$, Newey–West): negative, as predicted; the daily weight at average precision is 0.752, against 0.334 in HAR
- On most days the weight is near 0.76; on the noisiest days it falls (1% quantile 0.71, minimum 0.13): yesterday's RV is trusted only when it is precise
- Out of sample, QLIKE relative to HAR: Bitcoin 0.946 (DM -2.44), Ether 0.921 (DM -4.50): the gain of the original study carries over to crypto
- In MSE the picture differs (Bitcoin 1.163): MSE is dominated by a few extreme days, where HARQ shrinks the most; the AI mini-case checks how fragile the gain is

Realized GARCH (1/2): the three equations

- Hansen, Huang and Shek (2012), log-linear form: a return equation, a GARCH equation driven by the realised measure, and a measurement equation

$$r_t = \sqrt{h_t}\, z_t, \qquad \ln h_t = \omega + \beta \ln h_{t-\mathbf{1}} + \gamma \ln x_{t-\mathbf{1}}$$
$$\ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$$

  - h_t: conditional variance of the return r_t; z_t: i.i.d. standardised shock; x_t: realised measure of day t (here the realised kernel RK)
  - β: persistence of $\ln h_t$; γ: weight of yesterday's realised measure (the news variable that replaces $\varepsilon^{\mathbf{2}}_{t-\mathbf{1}}$)
  - u_t: measurement error, i.i.d. $N(0, \sigma_u^{\mathbf{2}})$, independent of z_t
- Measurement equation: x_t is a noisy, possibly biased signal of h_t
  - $\varphi = 1$: x_t proportional to h_t; ξ absorbs the scale (open-to-close measure against close-to-close variance)
  - leverage function $\tau(z) = \tau_{\mathbf{1}} z + \tau_{\mathbf{2}}(z^{\mathbf{2}} - 1)$: with $\tau_{\mathbf{1}} < 0$, negative returns raise the next realised measure (news impact, TSA, Chapter 5)

Realized GARCH (2/2): reduced form and estimation

- Substituting the measurement equation into the GARCH equation: $\ln h_t$ is an AR(1)

$$\ln h_t = (\omega + \gamma\xi) + \pi \ln h_{t-1} + \gamma\big(\tau(z_{t-1}) + u_{t-1}\big), \qquad \pi = \beta + \varphi\gamma$$

  - π: persistence of the log variance; multi-step forecasts follow, because x_t has its own equation
- Joint QML: the log-likelihood sums a return part and a measurement part

$$\ell = -\frac{1}{2}\sum_t \Big[\underbrace{\ln h_t + z_t^2}_{r_t} + \underbrace{\ln \sigma_u^2 + u_t^2/\sigma_u^2}_{x_t}\Big]$$

  - comparison with GARCH only through the partial likelihood of r_t (the first part), since GARCH does not model x_t

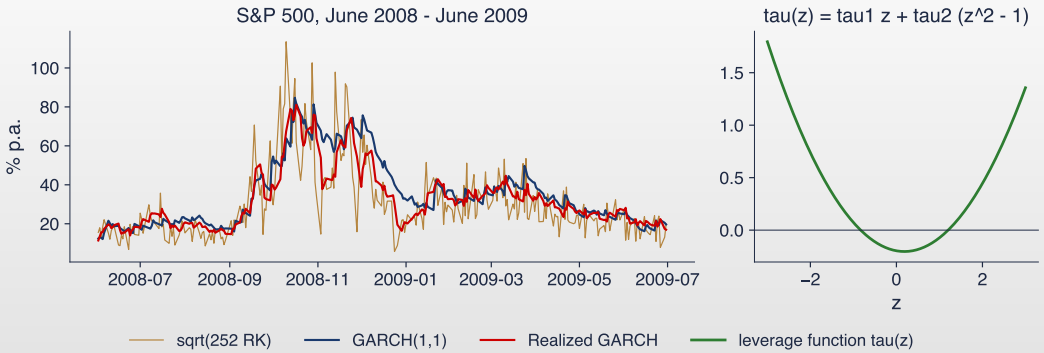
HEAVY and the family of realised-measure models

- HEAVY Shephard and Sheppard (2010): two GARCH-type equations, one for the return variance and one for the mean of the realised measure

$$h_t = \mathrm{Var}(r_t|\mathcal{F}_{t-\mathbf{1}}) = \omega + \alpha \mathrm{RM}_{t-\mathbf{1}} + \beta h_{t-\mathbf{1}}$$
$$\mu_t = \mathbb{E}(\mathrm{RM}_t|\mathcal{F}_{t-\mathbf{1}}) = \omega_R + \alpha_R \mathrm{RM}_{t-\mathbf{1}} + \beta_R \mu_{t-\mathbf{1}}$$

  - RM_t: realised measure of day t; (ω, α, β) and $(\omega_R, \alpha_R, \beta_R)$: the parameters of the two equations
  - each equation is estimated separately by QML; momentum: after a shock, h_t keeps rising while μ_t catches up
  - when the realised measure is in the variance equation, the squared return usually adds nothing ($\varepsilon^{\mathbf{2}}_{t-\mathbf{1}}$ gets a zero coefficient)
- Realized EGARCH Hansen and Huang (2016): several realised measures and a richer leverage; score-driven versions exist for fat tails
- Compared with HAR: returns and realised measures are modelled jointly, so densities and VaR/ES forecasts are available (Chapter 9)

Case study: Realized GARCH for the S&P 500



- Open-to-close returns and the Parzen realised kernel (Oxford-Man), 2000–2022, $T = 5\,552$, as in Hansen, Huang and Shek (2012); left: the 2008 crisis; right: the estimated leverage function

ATS_ch8_realized_garch

Realized GARCH estimates

ω	β	γ	ξ	φ	τ_1	τ_2	σ_u
0.119	0.678	0.287	−0.456	1.012	−0.073	0.197	0.671
(0.010)	(0.018)	(0.016)	(0.021)	(0.026)	(0.008)	(0.010)	(0.008)

- Bollerslev–Wooldridge s.e. in brackets; persistence $\hat{\pi} = \hat{\beta} + \hat{\varphi}\hat{\gamma} = 0.968$ (GARCH(1,1) on the same returns: 0.989)
- Partial log-likelihood of the returns: Realized GARCH −6899.7, HEAVY −6902.8, GJR −6998.8, GARCH(1,1) −7099.0
- HEAVY variance equation: $\hat{\alpha} = 0.377$ on RK_{t-1}, $\hat{\beta} = 0.691$

Interpreting Realized GARCH

- $\hat{\varphi} = 1.012$: the kernel is proportional to the conditional variance; $\hat{\xi} = -0.456$ with $\hat{\sigma}_u = 0.671$: on average the kernel lies below the conditional variance of the open-to-close return
- $\hat{\gamma} = 0.287$ is large and $\hat{\beta} = 0.678$ smaller than in GARCH: the realised measure brings information faster than squared returns (left panel, autumn 2008)
- Leverage: $\hat{\tau}_1 = -0.073$, $\hat{\tau}_2 = 0.197$: an asymmetric parabola, larger for negative z
- The gain in the partial likelihood over GARCH(1,1) (199.3 points) has the direction of the original paper; HEAVY is almost as good with three parameters

The proxy problem and robust losses (1/2)

- We never see σ_t^2: a forecast h_t is compared with a proxy $\hat{\sigma}_t^2$ (squared return, RV, RK)
  - the proxy is conditionally unbiased, $\mathbb{E}(\hat{\sigma}_t^2 | \mathcal{F}_{t-1}) = \sigma_t^2$, but noisy
- A loss L is *robust* Patton (2011) if the ranking of forecasts by $\mathbb{E}L(\hat{\sigma}_t^2, h_t)$ equals the ranking by $\mathbb{E}L(\sigma_t^2, h_t)$, for every unbiased proxy Hansen and Lunde (2006a)
- Necessary and sufficient condition: a Bregman form (Chapter 1, consistent scoring functions for the mean)

$$L(\hat{\sigma}^2, h) = \tilde{C}(h) + B(\hat{\sigma}^2) + C(h)(\hat{\sigma}^2 - h), \qquad C'(h) < 0$$

  - C: a decreasing function of the forecast; $\tilde{C}$: its antiderivative, $\tilde{C}' = C$; B: any function of the proxy alone (it does not affect the ranking)

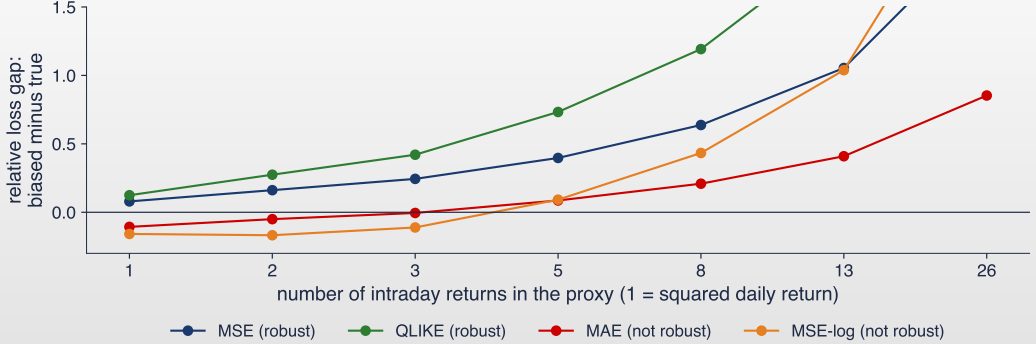
The proxy problem and robust losses (2/2)

- Robust: MSE and QLIKE, the members of degree 2 and degree 0 of the homogeneous robust family

$$\mathrm{MSE} = (\hat{\sigma}^2 - h)^2, \qquad \mathrm{QLIKE} = \frac{\hat{\sigma}^2}{h} - \ln\frac{\hat{\sigma}^2}{h} - 1$$

  - degree: how the loss scales when proxy and forecast are multiplied by the same constant; QLIKE (degree 0) is scale-free and penalises under-prediction more
- Not robust: MAE $= |\hat{\sigma}^2 - h|$, MSE on logs $(\ln\hat{\sigma}^2 - \ln h)^2$, MSE on standard deviations
  - with a noisy proxy they reward forecasts that are biased downwards
- Example: with $\hat{\sigma}^2 = r^2$, the MSE-log optimum is $h^* = \exp(\mathbb{E}\ln r^2) = 0.28\,\sigma^2$
  - because $r^2 = \sigma^2\chi_1^2$ for a Gaussian return and $\mathbb{E}\ln\chi_1^2 = -1.27$: MSE-log prefers a forecast far below the truth

Robust and non-robust losses



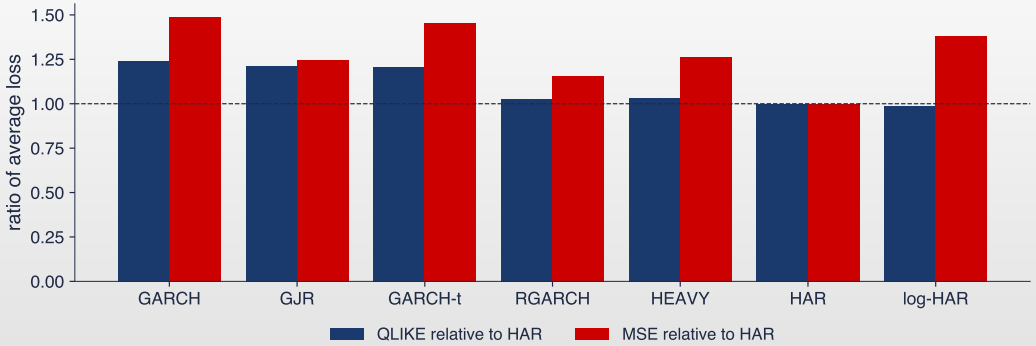
- True variance against a forecast biased down by the factor 0.6; proxy = RV from n intraday returns ($n = 1$: squared daily return); relative gap of expected losses, above zero = the true variance wins

ATS_ch8_robust_loss

Interpreting the robustness experiment

- With the squared return as proxy, MAE prefers the biased forecast (gap -10.7%) and so does MSE-log (-15.8%); MSE (7.9%) and QLIKE (12.5%) do not
- With $n = 5$ intraday returns all four losses rank correctly: a precise proxy makes non-robust losses "almost" robust Hansen and Lunde (2006a)
- The danger is largest exactly where realised measures are unavailable: daily data, emerging markets, long histories
- Rule: evaluate volatility forecasts with QLIKE (and MSE), against the best available proxy

Seven forecasts of S&P 500 volatility



- Open-to-close variance, one day ahead, 2016–2022 (1 537 days); expanding window from 2000, parameters re-estimated every 250 days; proxy: realised kernel

ATS_ch8_robust_loss

Interpreting the forecast comparison

- QLIKE: HAR 0.328, log-HAR 0.324, Realized GARCH 0.337, HEAVY 0.339, GARCH-t 0.396, GJR 0.398, GARCH 0.407
- DM against HAR (QLIKE): GARCH 8.62, GJR 6.27, Realized GARCH 1.42, log-HAR -0.37: realised-measure models beat return-only models clearly; among them the differences are small
- In MSE, HAR (1.60) leads and the DM statistics are smaller: the 2020 crash dominates the squared errors
- One proxy (RK) for all models: the HAR family is fitted to that same measure, which favours it; a fair comparison also tries another proxy (RV, BV)

From one variance to a covariance matrix (1/2)

- The vector of N return shocks is a matrix square root of the conditional covariance times a standardised vector

$$\varepsilon_t = H_t^{1/2}\eta_t, \qquad \eta_t \text{ i.i.d. } (0, I_N)$$

  - $H_t = \mathrm{Var}(\varepsilon_t|\mathcal{F}_{t-1})$: $N \times N$ conditional covariance matrix; I_N: identity matrix
  - H_t must be symmetric positive definite for every t and every parameter value
- VEC: every element of H_t depends on all past squares and cross-products

$$\mathrm{vech}(H_t) = c + A\,\mathrm{vech}(\varepsilon_{t-1}\varepsilon_{t-1}') + B\,\mathrm{vech}(H_{t-1})$$

  - vech: stacks the $N(N+1)/2$ distinct elements of a symmetric matrix; c: vector, A, B: square matrices of that size
  - general, but positivity is hard to impose

From one variance to a covariance matrix (2/2): BEKK, CCC and DCC

- BEKK Engle and Kroner (1995): quadratic forms guarantee positive definiteness

$$H_t = CC' + A'\varepsilon_{t-1}\varepsilon_{t-1}'A + B'H_{t-1}B$$

  - C: lower-triangular $N \times N$; A, B: $N \times N$ matrices; diagonal and scalar versions restrict A, B
  - identification: (A, B) and $(-A, -B)$ give the same H_t; fix the sign of one element

- CCC Bollerslev (1990) and DCC Engle (2002): variances from univariate GARCH, correlations modelled separately

$$H_t = D_t R D_t \ \text{(CCC)}, \qquad H_t = D_t R_t D_t \ \text{(DCC)}, \qquad D_t = \operatorname{diag}(\sigma_{1t}, \ldots, \sigma_{Nt})$$

  - σ_{it}: conditional standard deviation of asset i from its own GARCH; R: constant correlation matrix; R_t: time-varying correlation matrix

The curse of dimensionality in numbers

N	VEC	BEKK	diagonal BEKK	scalar BEKK	DCC
2	21	11	7	5	9
5	465	65	25	17	27
10	6 105	255	75	57	77
25	211 575	1 575	375	327	377
50	3 252 525	6 275	1 375	1 277	1 377
100	51 010 050	25 050	5 250	5 052	5 252

- Counts of free parameters of the (1,1) versions; DCC: $3N$ univariate GARCH parameters, 2 correlation parameters and the $N(N-1)/2$ elements of the target
- The DCC target is estimated by moments (correlation targeting), so the numerical search has only 2 parameters at any N; the price: $N(N-1)/2$ estimated correlations

Asymptotic theory for multivariate GARCH

- Gaussian QML is the standard estimator; consistency and asymptotic normality: VEC and BEKK Comte and Lieberman (2003), Hafner and Preminger (2009); CCC with ARMA means Ling and McAleer (2003)
- Conditions are stronger than in the univariate case (moments of order 6 or 8 for BEKK in early results); the sandwich of the univariate section carries over
- DCC is estimated in two steps Engle and Sheppard (2001): (1) univariate GARCH for each series; (2) QML of the correlation part given the standardised residuals $z_t = D_t^{-1}\varepsilon_t$
  - second-step standard errors must account for the first step (a GMM-type correction); naive second-step s.e. are too small
- Large N: the full likelihood needs R_t^{-1} and $|R_t|$ at each t ($O(N^3)$) and the score is dominated by noise; composite likelihood over pairs Pakel, Shephard, Sheppard and Engle (2021) avoids both

DCC and its corrected version cDCC (1/2): DCC



Robert F. Engle, 2017

Photo: Econterms (2017); CC BY-SA 4.0; Wikimedia Commons

- DCC: a GARCH(1,1)-type recursion for a quasi-correlation matrix Q_t, rescaled into a correlation matrix

$$Q_t = (1 - a - b)S + a\, z_{t-1} z_{t-1}' + b\, Q_{t-1}$$

$$R_t = \mathrm{diag}(Q_t)^{-1/2}\, Q_t\, \mathrm{diag}(Q_t)^{-1/2}$$

  - $z_t = D_t^{-1}\varepsilon_t$: standardised residuals of the univariate GARCH models; S: their sample correlation matrix (the target)
  - $a \geq 0$: reaction of correlations to yesterday's co-movement; $b \geq 0$: persistence; $a + b < 1$
  - $\mathrm{diag}(Q_t)$: the diagonal matrix of the diagonal elements of Q_t; the rescaling puts ones on the diagonal of R_t

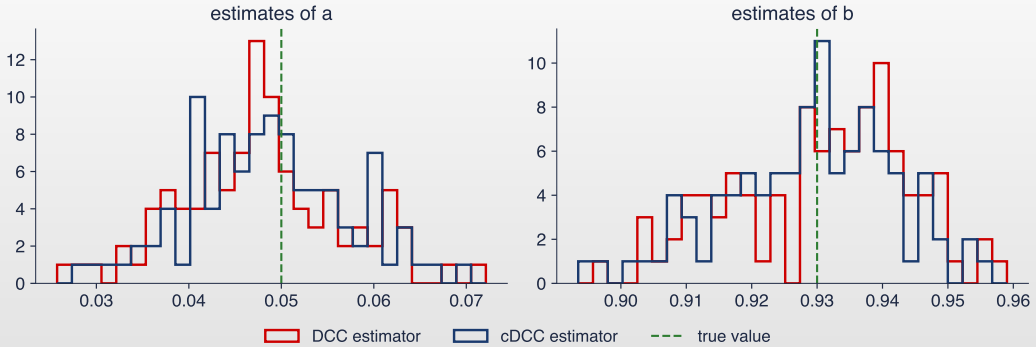
DCC and its corrected version cDCC (2/2): cDCC

- Aielli (2013): the DCC target is estimated inconsistently
  - $\mathbb{E}(z_t z_t' | \mathcal{F}_{t-1}) = R_t \neq Q_t$, so in general $\mathbb{E}Q_t \neq \mathbb{E}z_t z_t'$
  - the moment estimator of S is inconsistent, and so is $(\hat{a}, \hat{b})$
- cDCC: rescale the residuals by the diagonal of Q_{t-1} before they enter the recursion

$$z^*_{t-1} = \mathrm{diag}(Q_{t-1})^{1/2} z_{t-1}, \qquad Q_t = (1 - a - b)S + a\, z^*_{t-1} z^{*\prime}_{t-1} + b\, Q_{t-1}$$

  - then $\mathbb{E}(z^*_t z^{*\prime}_t | \mathcal{F}_{t-1}) = Q_t$, and $S = \mathbb{E}z^*_t z^{*\prime}_t$ is a valid target
  - the cDCC target depends on (a, b): it is estimated iteratively inside the likelihood

DCC against cDCC: a simulation



- Bivariate cDCC process with $a = 0.05$, $b = 0.93$, target correlation 0.5, unit variances; $T = 2\,000$, 100 replications; both estimators on each sample

ATS_ch8_mgarch

Interpreting the DCC simulation

- Mean estimates: DCC $\hat{a} = 0.0478$, $\hat{b} = 0.9307$; cDCC $\hat{a} = 0.0486$, $\hat{b} = 0.9288$ (true 0.05 and 0.93)
- RMSE of $\hat{a}$: 0.0090 (DCC) and 0.0086 (cDCC); of $\hat{b}$: 0.0135 and 0.0129: the inconsistency is real but small at these persistence levels
- Sampling error dominates the bias with two thousand observations; the difference matters more for the target in large systems and for inference on (a, b)
- A proof of inconsistency does not tell you the size of the bias: simulate it in your own design before choosing an estimator

Large covariance matrices: shrinkage and DCC-NL (1/2)

- With N/T not small, the eigenvalues of a sample covariance S are too dispersed
  - N: number of assets; T: number of observations; the largest eigenvalues are too large, the smallest too small; the inverse S^{-1} amplifies the error
- Global minimum-variance (GMV) portfolio: the weights with the smallest variance that sum to 1

$$w = \frac{\Sigma^{-1}1}{1'\Sigma^{-1}1}$$

  - Σ: covariance matrix of the returns; 1: vector of ones
  - it loads on the directions with the smallest, most underestimated eigenvalues
  - it depends only on Σ, so its realised risk measures the quality of the covariance forecast without noise from expected returns

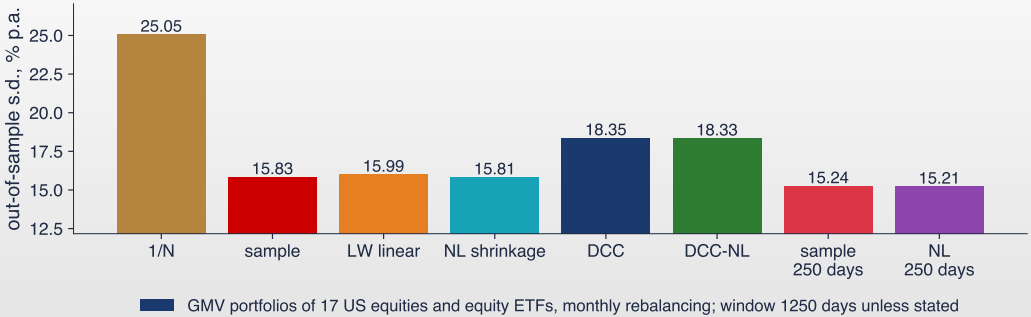
Large covariance matrices: shrinkage and DCC-NL (2/2)

- Linear shrinkage Ledoit and Wolf (2004): a weighted average of S and a scaled identity

$$\hat{\Sigma} = \delta\,\mu I + (1-\delta)\,S$$

  - μ: average eigenvalue of S; $\delta \in [0,1]$: shrinkage intensity, estimated from the data (larger when N/T is larger)
- Nonlinear shrinkage Ledoit and Wolf (2020): keep the eigenvectors of S, replace each eigenvalue λ_i by $d(\lambda_i)$
  - $d(\cdot)$: an analytical kernel formula that pulls small eigenvalues up and large ones down, by different amounts
- DCC-NL Engle, Ledoit and Wolf (2019): in DCC, the target S of the standardised residuals is itself a large sample covariance: replace it by its nonlinear shrinkage
  - composite likelihood for (a, b), univariate GARCH for the variances; evaluated by the out-of-sample s.d. of GMV portfolios

Case study: Engle, Ledoit and Wolf (2019) on US equities



- 17 assets: 14 US stocks and three equity ETFs (SPY, QQQ, RSP), which are portfolios of stocks; GMV weights rebalanced every 21 days from 23.05.2017 (112 rebalancings); window 1 250 days, and 250 days for the last two bars; annualised out-of-sample s.d.

ATS_ch8_mgarch

Interpreting the GMV comparison (1/2)

- Out-of-sample s.d. (% p.a.): 1/N 25.05, sample 15.83, linear shrinkage 15.99, nonlinear shrinkage 15.81, DCC 18.35, DCC-NL 18.33
- With $N/T = 17/1\,250$ the sample covariance is already accurate
  - nonlinear shrinkage changes little (15.81 against 15.83); DCC-NL is no better than DCC
- Here the dynamics hurt: DCC gives 18.35 against 15.83
  - DCC uses GARCH variances, with forecasts averaged over the 21-day holding period
  - one-month-ahead variance forecasts of single stocks such as MSTR and TSLA are noisy, and GMV amplifies their errors

Interpreting the GMV comparison (2/2)

- A window of 250 days makes N/T five times larger
  - s.d. of the sample GMV 15.24, with nonlinear shrinkage 15.21: only 0.2% lower
  - yet the ETFs make the correlation matrix nearly singular (median condition number 1146)
  - the shorter window itself lowers the risk: a crude form of time variation
- The original study uses hundreds of stocks, where N/T is large
  - there the DCC-NL gains are substantial
  - with our N the lesson is when shrinkage matters, not how much it gains

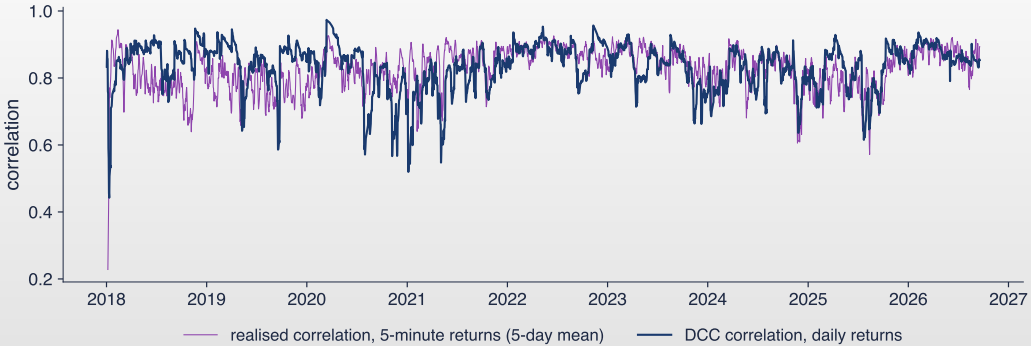
Realised covariance

- The multivariate RV: the sum of outer products of the intraday return vectors

$$\mathrm{RCov}_t = \sum_{i=1}^{n} r_{t,i}\, r_{t,i}' \to \int_0^1 \Sigma_s \, ds$$

  - $r_{t,i}$: $N \times 1$ vector of intraday returns; Σ_s: spot covariance matrix; the limit adds the co-jumps (simultaneous jumps); with synchronous, noise-free prices it inherits the CLT of RV
- Asynchronous trading: previous-tick prices create zero returns for the asset that has not traded; covariances shrink towards zero as the interval falls (the Epps effect)
  - refresh-time sampling (all assets have traded) and the multivariate realised kernel Barndorff-Nielsen et al. (2011): consistent and positive semi-definite
- Forecasting: HAR on the elements of the Cholesky factor of RCov_t keeps forecasts positive definite Chiriac and Voev (2011); HEAVY and Realized GARCH have multivariate versions
- Evaluating covariance forecasts H_t against a matrix proxy Σ_t: robust losses Laurent, Rombouts and Violante (2012)
  - matrix QLIKE $\ln|H_t| + \mathrm{tr}(H_t^{-1}\Sigma_t)$, with $|\cdot|$ the determinant and tr the trace; Frobenius MSE, the sum of squared element-wise errors

Bitcoin and Ether: realised and DCC correlation



- Daily realised correlation from 5-minute returns (5-day means) and the DCC correlation of daily close-to-close (UTC) returns, 2018–2026

ATS_ch8_mgarch

Interpreting the correlations

- Average realised correlation 0.83, average DCC correlation 0.83; correlation between the two series 0.26
- DCC uses one observation a day ($\hat{a} = 0.051$, $\hat{b} = 0.928$): it smooths and reacts with a lag; the realised measure moves daily and drops quickly in idiosyncratic episodes (5% quantile 0.68)
- Combining both (a realised measure in the correlation equation) is the multivariate analogue of Realized GARCH
- Five-minute sampling sits on the flat part of the Epps curve for these two coins (signature slide): the realised correlation is not biased towards zero

An open question

- Does exploiting the measurement error of realised variance (HARQ) improve volatility forecasts robustly, or only under particular design choices?
  - formal: the QLIKE ratio HARQ/HAR and its DM statistic across a pre-registered grid (window, filter, realised measure, asset); falsified if the ratio exceeds 1 or the DM test is insignificant for most cells
  - the measure of precision (RQ) is itself noisy and fat-tailed, so the gain may be fragile
- Why it matters: risk systems use one-day volatility forecasts every day; a small but robust gain is worth more than a large fragile one
  - literature to start from: Bollerslev, Patton and Quaedvlieg (2016), Corsi (2009), Patton (2011), Hansen and Lunde (2006a), Hansen, Lunde and Nason (2011)

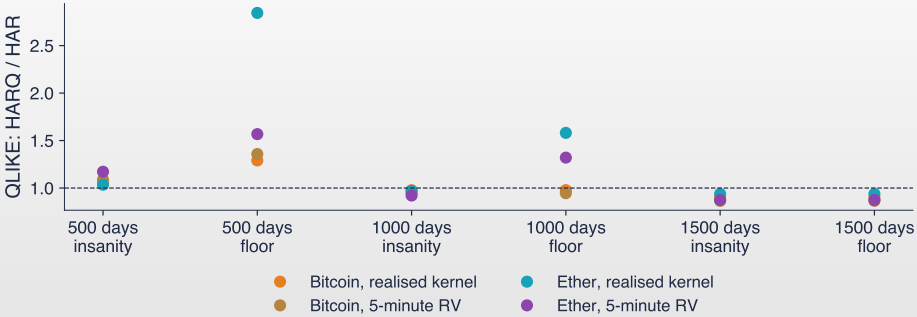
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers since 2016 that test HARQ-type models out of sample on assets other than US equities; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Under which data features (24/7 trading, jumps, zero returns) should the HARQ gain shrink or grow? Give testable predictions.`
  - **code and replication**: ask for HAR and HARQ rolling forecasts, then reproduce a known number first (the Bitcoin QLIKE ratio of this lecture)
  - **robustness and critique**: `Act as a hostile referee: which choices in a HARQ forecast comparison could produce a spurious gain?`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The forecast at t uses only data up to t: windows, demeaning of RQ, the insanity filter and the bias correction of log forecasts are all computed in the estimation window
- The loss is robust (QLIKE, MSE) and the proxy is stated; DM statistics use HAC variances
- All cells of the grid are reported, not the best one; the number of comparisons is stated
- Realised measures are recomputed from raw prices and agree with an independent source on overlapping assets and dates

Mini-case: robustness of the HARQ–HAR comparison



- QLIKE ratio HARQ/HAR for Bitcoin and Ether: window 500, 1 000 or 1 500 days; insanity filter or a floor at the smallest in-sample RV; target 5-minute RV or realised kernel
- Ratios from 0.865 to 2.844; HARQ better in 58% of the 24 cells, significantly better (DM < -1.96) in 50%, significantly worse in 12%: an AI summary reporting one cell would hide most of this

ATS_ch8_ai_robustness

Project idea

- **Realised volatility of crypto-assets: does the equity toolkit transfer?**: replicate first, then extend
  - replicate: HAR and HARQ of this lecture for Bitcoin (QLIKE ratio 0.946) and Realized GARCH for the S&P 500 (persistence 0.968)
  - extend: ten coins from the Binance archive; noise-robust measures from one-second data
  - extend the tests: an intraday seasonality correction before jump tests; Realized GARCH with the realised kernel
  - use the spot-ETF date (January 2024) as a natural break
  - pre-register: assets, sample, measures, models, windows, losses, the DM/MCS tests and the robustness grid
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- Gaussian QML needs only the variance equation; with fat tails, use sandwich standard errors
- Volatility has fast and slow components; macro drivers of the slow one need long samples and honest variance ratios
- Realised measures make volatility observable with a known error; noise and jumps decide the estimator
- HAR, HARQ and Realized GARCH beat return-only models one day ahead; evaluate with QLIKE or MSE
- In many dimensions the correlation target needs shrinkage (DCC-NL); whether the dynamics help must be checked out of sample

Self-assessment

Questions

- Why are Hessian standard errors too small for GARCH when $\kappa_\eta > 3$?
- What does the variance ratio of GARCH-MIDAS measure?
- Why does stable convergence matter for the CLT of RV?
- Why can a signature plot rise at high frequency?
- Why is β_{dQ} expected to be negative in HARQ?
- Which losses keep the ranking under a noisy proxy?

Next:

- VaR, ES and backtesting: elicitability, scoring and model risk
- From volatility forecasts to tail forecasts: VaR 1% and ES 2.5%, their scoring functions and backtests

Appendix: the GARCH sandwich

- $\ell_t = -\frac{1}{2}(\ln \sigma_t^2 + \varepsilon_t^2/\sigma_t^2)$, $d_t = \sigma_t^{-2}\partial\sigma_t^2/\partial\theta$: $s_t = \frac{1}{2}(\eta_t^2 - 1)d_t$ at θ_0
- $B = \mathbb{E}s_t s_t' = \frac{1}{4}\mathbb{E}(\eta_t^2 - 1)^2\,\mathbb{E}d_t d_t' = \frac{\kappa_\eta - 1}{4}J$, because η_t is independent of $d_t \in \mathcal{F}_{t-1}$
- $\partial s_t/\partial\theta' = -\frac{1}{2}\eta_t^2 d_t d_t' + \frac{1}{2}(\eta_t^2 - 1)\partial d_t/\partial\theta'$; the second term has mean zero, so $A = \frac{1}{2}J$
- $A^{-1}BA^{-1} = 4J^{-1} \cdot \frac{\kappa_\eta - 1}{4}J \cdot J^{-1} = (\kappa_\eta - 1)J^{-1}$; Hessian only: $A^{-1} = 2J^{-1}$; OPG: $B^{-1} = \frac{4}{\kappa_\eta - 1}J^{-1}$

Back

Appendix: the noise bias of realised variance

- $\tilde{r}_i = Y_{i/n} - Y_{(i-1)/n} = r_i + u_i - u_{i-1}$, with u i.i.d. $(0, \omega^2)$ independent of X
- $\sum_i \tilde{r}_i^2 = \sum_i r_i^2 + 2\sum_i r_i(u_i - u_{i-1}) + \sum_i (u_i - u_{i-1})^2$; the cross term has mean zero, the last has mean $2n\omega^2$
- $\mathrm{Cov}(\tilde{r}_i, \tilde{r}_{i-1}) = -\omega^2$: first-order negative autocorrelation; the realised kernel adds $2\gamma_1 \approx -2n\omega^2$ back and removes the bias
- Positively autocorrelated returns (gradual price adjustment, stale prices) give $\gamma_1 > 0$ and a signature plot that rises with the interval; the kernel corrects both signs

Back

Appendix: attenuation and HARQ

- $\mathrm{IV}_{t+1} = c + \phi \mathrm{IV}_t + v_{t+1}$, observed $\mathrm{RV}_t = \mathrm{IV}_t + e_t$, e_t uncorrelated with IV_t and v_{t+1}
- OLS of RV_{t+1} on RV_t: $\operatorname{plim} \hat{\phi} = \frac{\mathsf{Cov}(\mathrm{IV}_{t+1}, \mathrm{IV}_t)}{\mathsf{Var}(\mathrm{IV}_t) + \mathsf{Var}(e_t)} = \phi\lambda$, $\lambda = \frac{\mathsf{Var}(\mathrm{IV})}{\mathsf{Var}(\mathrm{IV}) + \mathbb{E}\mathsf{Var}(e_t)}$
- The best linear predictor given today's precision uses $\phi\lambda_t$, $\lambda_t = \frac{\mathsf{Var}(\mathrm{IV})}{\mathsf{Var}(\mathrm{IV}) + 2\mathrm{IQ}_t/n}$: decreasing in IQ_t
- A first-order expansion in $\sqrt{\mathrm{IQ}_t}$ gives $\beta_d + \beta_{dQ}\sqrt{\mathrm{RQ}_t}$ with $\beta_{dQ} < 0$: the HARQ specification Bollerslev, Patton and Quaedvlieg (2016)

Back

References (1/5)

- Aielli, G. P. (2013). Dynamic conditional correlation: On properties and estimation. *Journal of Business & Economic Statistics*, 31(3), 282–299. doi:10.1080/07350015.2013.771027
- Andersen, T. G., & Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905. doi:10.2307/2527343
- Andersen, T. G., Bollerslev, T., & Diebold, F. X. (2007). Roughing it up: Including jump components in the measurement, modeling, and forecasting of return volatility. *The Review of Economics and Statistics*, 89(4), 701–720. doi:10.1162/rest.89.4.701
- Andersen, T. G., Bollerslev, T., Diebold, F. X., & Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625. doi:10.1111/1468-0262.00418
- Andersen, T. G., Dobrev, D., & Schaumburg, E. (2012). Jump-robust volatility estimation using nearest neighbor truncation. *Journal of Econometrics*, 169(1), 75–93. doi:10.1016/j.jeconom.2012.01.011
- Bandi, F. M., & Russell, J. R. (2008). Microstructure noise, realized variance, and optimal sampling. *The Review of Economic Studies*, 75(2), 339–369. doi:10.1111/j.1467-937X.2008.00474.x
- Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2008). Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6), 1481–1536. doi:10.3982/ECTA6495
- Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2009). Realized kernels in practice: Trades and quotes. *The Econometrics Journal*, 12(3), C1–C32. doi:10.1111/j.1368-423X.2008.00275.x
- Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., & Shephard, N. (2011). Multivariate realised kernels: Consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2), 149–169. doi:10.1016/j.jeconom.2010.07.009
- Barndorff-Nielsen, O. E., & Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B*, 64(2), 253–280. doi:10.1111/1467-9868.00336
- Barndorff-Nielsen, O. E., & Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1), 1–37. doi:10.1093/jjfinec/nbh001
- Barndorff-Nielsen, O. E., & Shephard, N. (2006). Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, 4(1), 1–30. doi:10.1093/jjfinec/nbi022

References (2/5)

- Berkes, I., Horváth, L., & Kokoszka, P. (2003). GARCH processes: Structure and estimation. *Bernoulli*, 9(2), 201–227. doi:10.3150/bj/1068128975
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. doi:10.1016/0304-4076(86)90063-1
- Bollerslev, T. (1990). Modelling the coherence in short-run nominal exchange rates: A multivariate generalized ARCH model. *The Review of Economics and Statistics*, 72(3), 498–505. doi:10.2307/2109358
- Bollerslev, T., Patton, A. J., & Quaedvlieg, R. (2016). Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1), 1–18. doi:10.1016/j.jeconom.2015.10.007
- Bollerslev, T., & Wooldridge, J. M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172. doi:10.1080/07474939208800229
- Chiriac, R., & Voev, V. (2011). Modelling and forecasting multivariate realized volatility. *Journal of Applied Econometrics*, 26(6), 922–947. doi:10.1002/jae.1152
- Comte, F., & Lieberman, O. (2003). Asymptotic theory for multivariate GARCH processes. *Journal of Multivariate Analysis*, 84(1), 61–84. doi:10.1016/S0047-259X(02)00009-X
- Conrad, C., & Kleen, O. (2020). Two are better than one: Volatility forecasting using multiplicative component GARCH-MIDAS models. *Journal of Applied Econometrics*, 35(1), 19–45. doi:10.1002/jae.2742
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196. doi:10.1093/jjfinec/nbp001
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- Engle, R. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3), 339–350. doi:10.1198/073500102288618487
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007. doi:10.2307/1912773

References (3/5)

- Engle, R. F., Ghysels, E., & Sohn, B. (2013). Stock market volatility and macroeconomic fundamentals. *The Review of Economics and Statistics*, 95(3), 776–797. doi:10.1162/REST_a_00300
- Engle, R. F., & Kroner, K. F. (1995). Multivariate simultaneous generalized ARCH. *Econometric Theory*, 11(1), 122–150. doi:10.1017/S0266466600009063
- Engle, R. F., Ledoit, O., & Wolf, M. (2019). Large dynamic covariance matrices. *Journal of Business & Economic Statistics*, 37(2), 363–375. doi:10.1080/07350015.2017.1345683
- Engle, R. F., & Lee, G. G. J. (1999). A long-run and short-run component model of stock return volatility. In R. F. Engle & H. White (Eds.), *Cointegration, Causality, and Forecasting: A Festschrift in Honour of Clive W. J. Granger* (pp. 475–497). Oxford University Press. doi:10.1093/oso/9780198296836.003.0020
- Engle, R. F., & Sheppard, K. (2001). Theoretical and empirical properties of dynamic conditional correlation multivariate GARCH. *NBER Working Paper* 8554. doi:10.3386/w8554
- Epps, T. W. (1979). Comovements in stock prices in the very short run. *Journal of the American Statistical Association*, 74(366), 291–298. doi:10.1080/01621459.1979.10482508
- Francq, C., & Zakoïan, J.-M. (2004). Maximum likelihood estimation of pure GARCH and ARMA-GARCH processes. *Bernoulli*, 10(4), 605–637. doi:10.3150/bj/1093265632
- Francq, C., & Zakoïan, J.-M. (2019). *GARCH Models: Structure, Statistical Inference and Financial Applications* (2nd ed.). Wiley. doi:10.1002/9781119313472
- Ghysels, E., Santa-Clara, P., & Valkanov, R. (2006). Predicting volatility: Getting the most out of return data sampled at different frequencies. *Journal of Econometrics*, 131(1–2), 59–95. doi:10.1016/j.jeconom.2005.01.004
- Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5), 1779–1801. doi:10.1111/j.1540-6261.1993.tb05128.x
- Hafner, C. M., & Preminger, A. (2009). On asymptotic theory for multivariate GARCH models. *Journal of Multivariate Analysis*, 100(9), 2044–2054. doi:10.1016/j.jmva.2009.03.011
- Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. doi:10.2307/j.ctv14jx6sm

References (4/5)

- Hansen, P. R., & Huang, Z. (2016). Exponential GARCH modeling with realized measures of volatility. *Journal of Business & Economic Statistics*, 34(2), 269–287. doi:10.1080/07350015.2015.1038543
- Hansen, P. R., Huang, Z., & Shek, H. H. (2012). Realized GARCH: A joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6), 877–906. doi:10.1002/jae.1234
- Hansen, P. R., & Lunde, A. (2005). A forecast comparison of volatility models: Does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7), 873–889. doi:10.1002/jae.800
- Hansen, P. R., & Lunde, A. (2006a). Consistent ranking of volatility models. *Journal of Econometrics*, 131(1–2), 97–121. doi:10.1016/j.jeconom.2005.01.005
- Hansen, P. R., & Lunde, A. (2006b). Realized variance and market microstructure noise. *Journal of Business & Economic Statistics*, 24(2), 127–161. doi:10.1198/073500106000000071
- Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497. doi:10.3982/ECTA5771
- Heber, G., Lunde, A., Shephard, N., & Sheppard, K. (2009). *Oxford-Man Institute's realized library*, version 0.3. Oxford-Man Institute, University of Oxford (discontinued in 2022; copy of 28 February 2022 preserved by the Internet Archive). web.archive.org
- Huang, X., & Tauchen, G. (2005). The relative contribution of jumps to total price variance. *Journal of Financial Econometrics*, 3(4), 456–499. doi:10.1093/jjfinec/nbi025
- Jacod, J., Li, Y., Mykland, P. A., Podolskij, M., & Vetter, M. (2009). Microstructure noise in the continuous case: The pre-averaging approach. *Stochastic Processes and their Applications*, 119(7), 2249–2276. doi:10.1016/j.spa.2008.11.004
- Jensen, S. T., & Rahbek, A. (2004). Asymptotic inference for nonstationary GARCH. *Econometric Theory*, 20(6), 1203–1226. doi:10.1017/S0266466604206065
- Laurent, S., Rombouts, J. V. K., & Violante, F. (2012). On the forecasting accuracy of multivariate GARCH models. *Journal of Applied Econometrics*, 27(6), 934–955. doi:10.1002/jae.1248
- Ledoit, O., & Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411. doi:10.1016/S0047-259X(03)00096-4

References (5/5)

- Ledoit, O., & Wolf, M. (2020). Analytical nonlinear shrinkage of large-dimensional covariance matrices. *The Annals of Statistics*, 48(5), 3043–3065. doi:10.1214/19-AOS1921
- Lee, S. S., & Mykland, P. A. (2008). Jumps in financial markets: A new nonparametric test and jump dynamics. *The Review of Financial Studies*, 21(6), 2535–2563. doi:10.1093/rfs/hhm056
- Lee, S.-W., & Hansen, B. E. (1994). Asymptotic theory for the GARCH(1,1) quasi-maximum likelihood estimator. *Econometric Theory*, 10(1), 29–52. doi:10.1017/S0266466600008215
- Ling, S., & McAleer, M. (2003). Asymptotic theory for a vector ARMA-GARCH model. *Econometric Theory*, 19(2), 280–310. doi:10.1017/S0266466603192092
- Lumsdaine, R. L. (1996). Consistency and asymptotic normality of the quasi-maximum likelihood estimator in IGARCH(1,1) and covariance stationary GARCH(1,1) models. *Econometrica*, 64(3), 575–596. doi:10.2307/2171862
- Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. doi:10.2307/1913610
- Pakel, C., Shephard, N., Sheppard, K., & Engle, R. F. (2021). Fitting vast dimensional time-varying covariance models. *Journal of Business & Economic Statistics*, 39(3), 652–668. doi:10.1080/07350015.2020.1713795
- Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256. doi:10.1016/j.jeconom.2010.03.034
- Patton, A. J., & Sheppard, K. (2015). Good volatility, bad volatility: Signed jumps and the persistence of volatility. *The Review of Economics and Statistics*, 97(3), 683–697. doi:10.1162/REST_a_00503
- Petropoulos, F., Apiletti, D., Assimakopoulos, V., Babai, M. Z., et al. (2022). Forecasting: Theory and practice. *International Journal of Forecasting*, 38(3), 705–871. doi:10.1016/j.ijforecast.2021.11.001
- Shephard, N., & Sheppard, K. (2010). Realising the future: Forecasting with high-frequency-based volatility (HEAVY) models. *Journal of Applied Econometrics*, 25(2), 197–231. doi:10.1002/jae.1158
- Zhang, L., Mykland, P. A., & Aït-Sahalia, Y. (2005). A tale of two time scales: Determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472), 1394–1411. doi:10.1198/016214505000000169